Mapúa University

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MapúOne: Smart Campus Complaint & Case Management System for Mapúa University

A Thesis Paper Presented to

School of Information Technology

Mapúa University - Makati

In Partial Fulfillment of the Requirements for the Course

ITS-200

3rd Term / 2025-2026

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May 2026

TABLE OF CONTENTS

CHAPTER 1 - INTRODUCTION

Efficient management of complaints and concerns is critical to maintaining a responsive, student-centered learning environment. As universities continue to grow in size and operational complexity, traditional grievance-handling methods—often reliant on informal communication channels, paper-based documentation, and fragmented administrative processes—have become increasingly inadequate. These conventional approaches frequently lead to delayed resolutions, misplaced records, a lack of accountability, and diminished stakeholder confidence in institutional responsiveness.

Mapúa University, a leading technological institution in the Philippines, serves a diverse community of students, faculty, and staff whose concerns span multiple domains, including academic affairs, facilities management, technical support, and student services. Currently, the absence of a unified digital platform for complaint submission and case tracking poses significant challenges to operational efficiency and transparency. Stakeholders often lack visibility into the status of their submitted concerns, while administrators struggle to monitor case progress, identify systemic issues, and generate actionable insights from complaint data.

In response to these challenges, this study proposes developing a **Smart Campus Complaint & Case Management System** to centralize, streamline, and digitize the entire complaint lifecycle within the university. The proposed system will enable users to submit categorized complaints with supporting documentation, track real-time case status updates, and receive timely notifications about resolution progress. Administrators will also benefit from a comprehensive dashboard that facilitates case assignment, rule-based complaint routing, status management, and basic administrative reporting.

This research seeks to address the following problems:

1. Inefficient manual complaint-handling processes that create administrative delays
2. The absence of a centralized repository for complaint records leads to fragmented documentation
3. Prolonged resolution timelines due to inadequate case-tracking mechanisms
4. Limited transparency for complainants regarding the progress and outcome of their submissions
5. The lack of a centralized tracking system to monitor case progress, identify recurring issues, and support basic administrative reporting on campus complaint resolution

By implementing this system, the study aims to enhance institutional accountability, improve service delivery, and foster a more transparent and responsive campus environment at Mapúa University.

1.1 Background of the study

In the contemporary landscape of higher education, institutional responsiveness and operational efficiency have become essential components of effective university governance. As academic institutions grow in size and complexity, the need for reliable systems that address stakeholder concerns promptly and transparently has never been more critical. Complaint management, in particular, plays a vital role in maintaining trust between the administration and its community of students, faculty, and staff. When grievances are handled effectively, institutions not only resolve individual concerns but also gain valuable insight into systemic issues that can inform broader improvements in campus services and policies.

Despite its importance, many universities — including those in the Philippines — continue to rely on outdated, manual methods of handling grievances. These approaches, characterized by paper-based documentation, informal communication channels, and fragmented administrative processes, frequently result in delayed resolutions, misplaced records, and limited accountability. Complainants are often left uninformed about the status of their submissions, while administrators struggle to monitor case progress or identify recurring institutional issues. As universities expand and stakeholder expectations rise, these conventional methods have become increasingly inadequate in meeting the demands of a modern academic environment.

Within the Philippine context, the Commission on Higher Education (CHED) and various accrediting bodies have increasingly emphasized the importance of institutional accountability, stakeholder satisfaction, and evidence-based governance as benchmarks of academic quality. In this environment, the absence of a centralized, digitized complaint management platform represents not merely an operational gap but also a governance concern that directly affects the quality of service delivery within higher education institutions.

Mapúa University, a leading technological institution in the Philippines, is not exempt from these challenges. Despite its reputation for innovation and academic rigor, the university currently lacks a unified digital platform for complaint submission and case tracking. Stakeholders navigating multiple offices without clear visibility into the progress of their concerns often feel frustrated and lose confidence in the institution's responsiveness. Meanwhile, administrators are left without the tools necessary to manage cases systematically, ensure timely resolution, or derive meaningful insights from complaint data.

In response to these gaps, this study proposes the development of a Smart Campus Complaint and Case Management System. The system seeks to centralize and digitize the entire complaint lifecycle, enabling users to submit categorized concerns with supporting documentation, track case status in real time, and receive timely notifications on resolution progress. Administrators will benefit from a comprehensive dashboard for case assignment, rule-based complaint routing, progress monitoring, and basic administrative reporting. Through this initiative, the study aims to strengthen institutional accountability, improve service delivery, and foster a more responsive and transparent campus environment at Mapúa University.

1.2 Reference Studies

Several existing studies on digital complaint and grievance management systems in academic and institutional settings serve as direct references for the development of the proposed Smart Campus Complaint and Case Management System for Mapúa University.

Reddy et al. (2026) developed CampusFix, a smart campus complaint management system that uses machine learning to automatically categorize and route complaints to the appropriate department within a university setting. The system demonstrates how automated classification can significantly reduce response times and minimize misdirected complaints, providing a useful reference for the routing and case assignment features of the proposed system.

Alifat et al. (2022) presented the Web-based Campus Complaint Management System (WCCMS), which centralized complaint submission and automated the routing of concerns to appropriate departments while providing users with a transparent mechanism for monitoring submission progress. Their work directly informs the complaint submission and real-time tracking modules of the proposed system, demonstrating the viability of a web-based approach to complaint management in higher education.

Oguntosin et al. (2021) developed a web-based complaint management platform for a university community built using JavaScript and a MongoDB database, enabling complaints to be submitted, tracked, and monitored through a single centralized application. Their emphasis on centralized record management, role-based access, and administrator case oversight serves as a practical reference for the administrative dashboard and case management features of the proposed system.

Bhange et al. (2026) proposed a Smart Complaint Management System that digitalized the entire complaint lifecycle through a centralized web-based platform with role-based authentication, detailed complaint forms, status tracking across stages such as Pending, In Progress, and Resolved, and built-in reporting dashboards. Their work reinforces the importance of structured status workflows and administrative performance monitoring, both of which are incorporated into the proposed system.

Goyal et al. (2025) discussed a student review and complaint system designed to bridge the communication gap between students and governance bodies in higher education through real-time tracking dashboards and centralized digital logs accessible to department heads and administrative committees. Their system highlights how organized complaint records support departmental performance assessment and recurring issue identification, informing the basic reporting features of the proposed system.

Priyanka and Mahesh (2025) developed a grievance management system that streamlines complaint resolution through a centralized platform offering real-time complaint tracking, organized record management, and structured communication between students and administrators. Their study reinforces the value of transparency and accessibility in complaint systems and supports the notification and status tracking features included in the proposed system.

Al-Waeli and Hassan (2022) presented a Student Complaint Management System designed as a structured web-based platform that enables direct collection of student complaints, maps them to service metrics, and dynamically tracks them through resolution. Their emphasis on audit histories, structured case records, and scalability provides a relevant reference for the centralized database and case history management components of the proposed system.

Patle et al. (2023) presented a campus complaint management system featuring secure authentication, detailed complaint forms, real-time status tracking, notification support, and an administrative dashboard for monitoring complaint trends and resolution times. Their work serves as a direct reference for the integrated design of user-facing and administrator-facing interfaces in the proposed system.

Rahman and Mustaffar (2026) evaluated the e-Aduan system at Universiti Teknologi MARA, a centralized digital complaint platform that demonstrated how automated tracking pipelines and web-based administrative dashboards can support operational excellence and institutional quality assurance in a large university environment. Their findings validate the approach taken by the proposed system in centralizing complaint records and providing administrators with organized case management tools.

Srivastava et al. (2025) proposed a decision support complaint prioritization system that uses text mining and sentiment analysis to automatically assign priority values to incoming complaints, enabling administrators to manage records systematically through a secure management dashboard. While their focus on intelligent prioritization goes beyond the current scope of the proposed system, their work informs the future development direction of the platform and reinforces the importance of structured data pipelines in complaint management.

Collectively, these reference studies establish the foundational principles that guide the design and development of the proposed Smart Campus Complaint and Case Management System for Mapúa University. They consistently demonstrate that centralized, web-based complaint platforms improve institutional transparency, reduce administrative delays, and strengthen communication between complainants and university administrators. The proposed system builds upon these established foundations by integrating complaint submission, rule-based category routing, real-time status tracking, in-system and email notifications, and basic administrative reporting into a unified platform tailored to the specific operational and institutional requirements of a Philippine technological university.

1.3 Gap Analysis

A review of the available literature shows that although there has been significant development of digital grievance tools for academic settings, no comprehensive framework has been developed on the local level that includes structured, multi-role complaint management with category-based routing and multi-stage administrative case processing pipelines specifically tailored for a complex Philippine technological university. Although previous research offers examples of specific functional components to be used as a point of reference, it often showcases isolated forms of segmentation that are not applicable in the real world in the context of the structural requirements of Mapúa University.

The dichotomy between complaint routing processes and in-depth case resolution processes in existing software architectures is most evident. While advanced systems such as CampusFix (Reddy et al., 2026), the prioritization prototype (Srivastava et al., 2025), and the data-mining support system (Oliver et al., 2025) demonstrate promising capabilities in automated complaint classification and intelligent triage, the vast majority of them are limited solely to front-end processing tasks. These architectures are effective for initial categorization, but they do not provide the multi-stage, longitudinal tracking pipelines necessary for administrative offices to execute, update, and close tickets horizontally across other university departments. On the other hand, conventional web applications developed by Bhange et al. (2026), Goyal et al. (2025), and Oguntosin et al. (2021) offer organized role-based tracking portals with defined status stages such as Pending, In Progress, and Resolved; however, they rely on entirely manual routing setups. This architectural constraint requires that university staff manually review each submission to determine the appropriate domain, priority, and administrative destination for individual complaints, creating unnecessary delays in the resolution process.

In addition, current grievance systems tend to treat complaint handling as a linear process, where ticket closure is seen as the termination of a case rather than as an opportunity for institutional learning. Most existing systems do not maintain accessible resolution histories, departmental turnaround records, or organized case outcome logs that administrators can use to identify recurring institutional problems or areas requiring service improvement. This absence of structured complaint records limits the ability of institutions to make evidence-based decisions about campus service delivery.

Furthermore, foreign architectures have largely failed to adapt to the socio-political and institutional context of Philippine higher education environments. Existing literature describes systems designed for single-campus or homogeneous administrative structures with a general complaint queue managed from one central location. These systems do not reflect the dense, decentralized administrative setup of local institutions such as Mapúa University, where concerns are organized across specialized offices, including Facilities Management, Academic Affairs, IT/Technical Support, and Student Services. Moreover, most existing systems are designed to serve only students, excluding faculty and staff who may also need to raise campus-related concerns through a structured digital channel. Conventional open-source applications also lack built-in constraints for data validation suited to local governance requirements and routing workflows that verify case assignment across multiple offices.

The proposed Smart Campus Complaint & Case Management System for Mapúa University directly addresses this gap by providing a structured, rule-based routing mechanism that automatically directs complaints to the appropriate department based on complaint category, eliminating the need for manual triage. Unlike AI-driven systems that prioritize intelligent classification but lack multi-stage case resolution workflows, the proposed system integrates complaint submission, automated category-based routing, real-time tracking, notifications, and basic administrative reporting into a unified platform designed for the decentralized administrative structure of Mapúa University. The system maintains a centralized complaint history that supports basic administrative reporting and serves as a foundation for future data-driven enhancements. Advanced features such as machine learning classification, severity prediction, and predictive analytics are identified as viable future enhancements once the system has accumulated sufficient complaint records from actual university use, consistent with the developmental trajectory demonstrated in the existing literature. This results in a focused, institutionally aligned platform designed to enhance service transparency, reduce resolution delays, and improve the handling of campus-related concerns within Mapúa University.

1.4 Research Question

This study developed a Web-Based Smart Campus Complaint and Case Management System to address the inefficiencies of the current manual grievance handling processes at Mapúa University. Specifically, it sought to answer the question: How can a web-based complaint and case management system improve the submission, rule-based routing, real-time tracking, and resolution of campus-related concerns at Mapúa University compared to current manual processes, and how do end-users evaluate the system in terms of usability, functionality, and efficiency?

1.5 Objective Of The Study

A. General Objective

The general objective of this study is to design, develop, and evaluate a web-based Smart Campus Complaint and Case Management System (SCCCMS-MU) for Mapúa University—Makati that centralizes and digitizes the full complaint lifecycle—from structured submission and rule-based routing through multi-stage status tracking and resolution—for use by students, faculty, and staff as complainants and by authorized university administrators as case handlers, within the 3rd Term of Academic Year 2025–2026 and 1st Term of Academic Year 2026–2027.

B. Specific Objectives

1. To develop a web-based complaint submission module enabling students, faculty, and staff to register, log in, and submit categorized complaints with supporting attachments.

2. To develop a rule-based routing mechanism that automatically pre-assigns complaints to the appropriate department based on keyword detection, subject to administrator confirmation or override.

3. To develop a multi-stage case tracking system that provides complainants with real-time status visibility and notifications throughout the resolution process.

4. To develop an administrative dashboard enabling authorized personnel to manage, route, escalate, and resolve complaints.

5. To establish a centralized repository of resolved cases that generates administrative reports on complaint volume, resolution timelines, and departmental activity.

6. To evaluate the system's usability and functionality through structured testing with student and administrator respondents using a standardized evaluation instrument.

1.6 Existing Process Flow

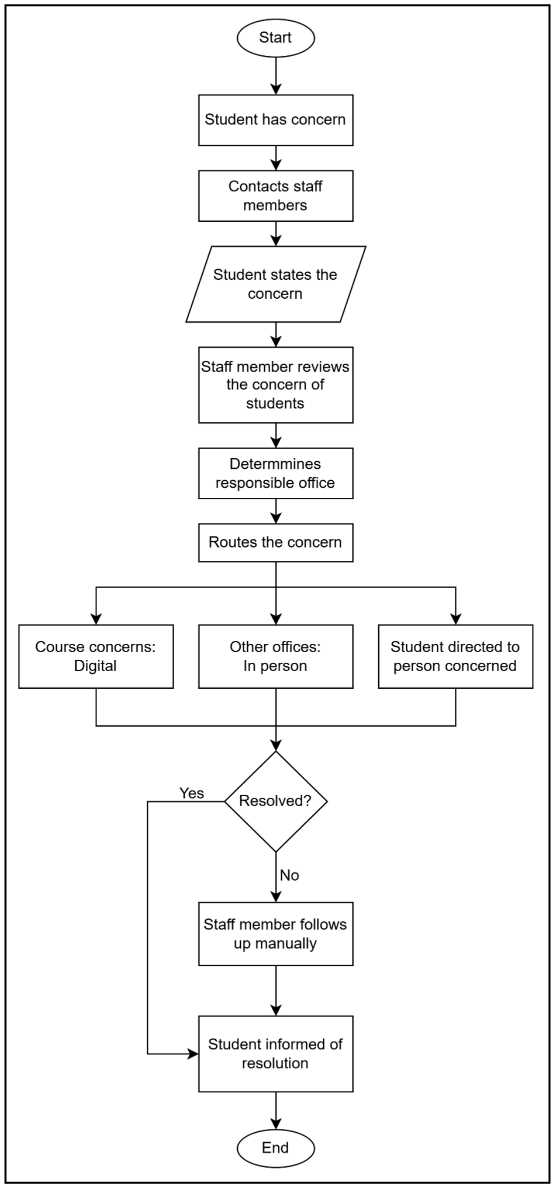

Figure 1: Existing System

The figure illustrates the existing manual complaint-handling process at Mapúa University, based on the accounts of two personnel who directly receive and handle student concerns: the Student Success Coordinator of the Institute for Digital Learning (IDL) and a Life Coach who assists online students. At present, the university does not employ a centralized digital platform for managing student concerns; instead, complaints are received and processed through informal, communication-based channels that depend heavily on the direct involvement and personal follow-through of individual staff members.

The process begins when a student encounters a concern, which may involve course-related matters, communication issues with facilitators, technical difficulties, enrollment, registrar transactions, or requests for academic support. Rather than submitting a formal complaint through a dedicated office or system, the student contacts a staff member directly. The coordinator prefers to be reached through Microsoft Teams, with Outlook email as an alternative, while the Life Coach asks online students to send their concerns through email so that these can be followed up. In both cases, there is no official complaint form to accomplish, and the message itself serves as the complaint. Students are expected to identify themselves: the coordinator requires the full name, course, and section along with a clear description of the concern, while the Life Coach receives the student's identity through the email itself. The volume handled is considerable, with the coordinator estimating that she receives at least thirty concerns on average and the Life Coach more than twenty per month. Since each concern arrives as a personal message, it is visible only to the staff member who received it, and there is no shared queue accessible to other offices.

Once the concern is received, the staff member reviews and assesses it to determine its nature and the appropriate course of action. There is, however, no single written policy governing how concerns should be ranked. The coordinator addresses concerns on a first-come, first-served basis, following the order in which students made contact, while the Life Coach prioritizes according to urgency and impact, attending first to safety issues, urgent facilities problems, and matters that affect many students. Both accounts indicate that this judgment rests largely on the individual handler, guided only in part by general guidelines and shaped mostly by personal experience.

The concern is then routed depending on its subject. Course-related concerns are generally referred digitally through email or Microsoft Teams to the relevant faculty or facilitators. When a concern involves other offices, such as the Treasury or Registrar, the coordinator addresses the matter personally by visiting the respective office, allowing her to discuss the concern directly and facilitate a faster resolution. The Life Coach, on the other hand, advises the student to message the specific person concerned, such as the professor, while being included in the email for monitoring purposes, and refers difficulties encountered by online students to the IDL. In cases where a concern involves more than one office or where responsibility is unclear, the coordinator reviews the details, identifies all the areas involved, and coordinates with the appropriate offices to ensure the concern is directed correctly.

If the concern can be resolved at the level of the assigned office or personnel, it is addressed accordingly, and a concern is generally treated as resolved once the student receives the answer needed and requests no further follow-up. No prior approval from a higher authority is required for routine concerns. However, when an issue cannot be resolved at that level, it is escalated to a higher authority such as a department head, director, or dean, and when a concern remains pending without a response from the person needed, the student may also be advised to include the heads in the email thread. Throughout this stage, the staff member who received the concern remains responsible for following up, monitoring its progress, and reminding the responsible office or personnel when necessary. Both respondents identified delays as a recurring problem, occurring when a concern is overlooked, when action is awaited from another department, or, as the Life Coach noted, when faculty members are slow to respond to matters such as student activities and delayed tasks.

The student is informed that the concern has been received through an acknowledgment from the staff member or the concerned department, and later that it has been resolved, through Microsoft Teams, Outlook email, or a phone call. Progress updates in between vary by handler: the coordinator noted that students are not regularly updated and that she instead takes the initiative to follow up on pending concerns, while the Life Coach updates students through email, phone, or official messages when there are significant developments. If additional information or clarification is needed, the student is contacted through the same channels.

Throughout the process, tracking is maintained informally through email threads, which serve as the only record of the communication and updates related to a concern, with cases simply remarked as resolved or still pending. The coordinator confirmed that there is no means of determining the current status of a complaint at any given moment. After resolution, there is no formal long-term record retention process in place, as resolved concerns are not systematically archived or maintained as permanent records, even though both respondents observed that many of the problems raised are recurring and must be resolved again as they reappear.

This existing process reveals several limitations that the proposed system seeks to address, including the absence of a standardized submission channel, the dependence on individual staff members for receiving and following up on concerns, the lack of real-time status visibility for students, the inconsistent handling of prioritization and progress updates between offices, the reliance on informal email-based tracking, and the absence of a structured record-keeping mechanism. Delays in resolution, particularly when action is required from other offices or from faculty, were identified by both respondents as the most significant challenge under the current setup.

1.7 Proposed Process Flow

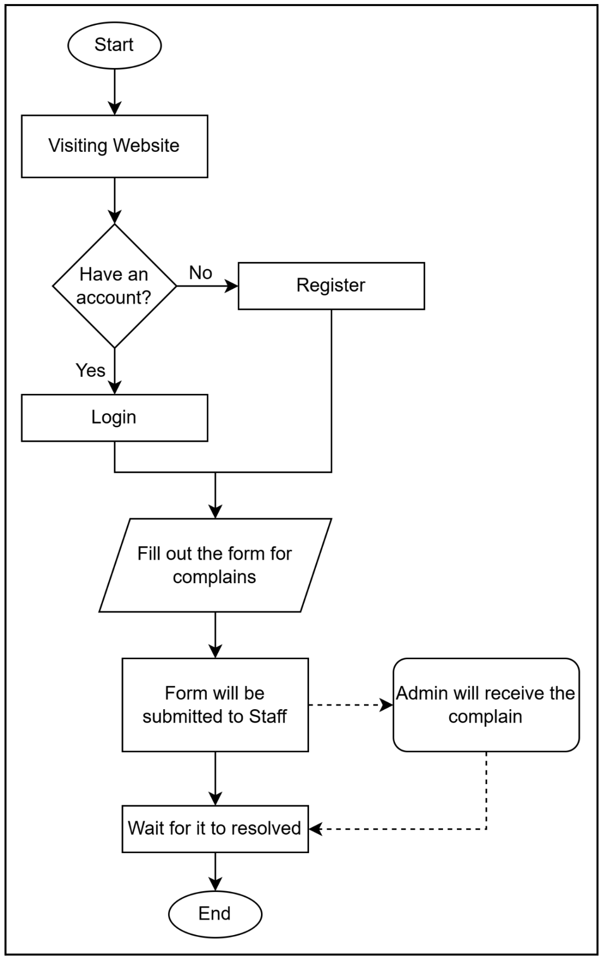

Figure 2: User Process Flow

The following figure illustrates the proposed user process flow of the system. When a user decides to report a complaint or concern related to school matters, they will first access the platform. If the user does not yet have an account, they will be prompted to register by providing their personal information in order to access and track all submitted cases.

After logging in, the user will complete a complaint form by selecting the appropriate category, providing the necessary details of their concern, and optionally attaching supporting files for additional context. Once submitted, the system automatically generates a case record, assigns it an initial status of "Open," and pre-routes the complaint to the relevant department based on its category before forwarding it to the administrator for review and processing.

The user can then monitor the status of their case through a progress tracker within the system, which displays each stage of handling such as Open, In Process, Pending Response, and Resolved. Updates and comments from the assigned personnel will also be visible to keep the user informed about the case development, and the user may receive in-system or email notifications when the status changes. If needed, the user may communicate with the assigned staff through the system to provide additional information or clarification regarding the complaint until the case is resolved.

1.8 Admin Process Flow

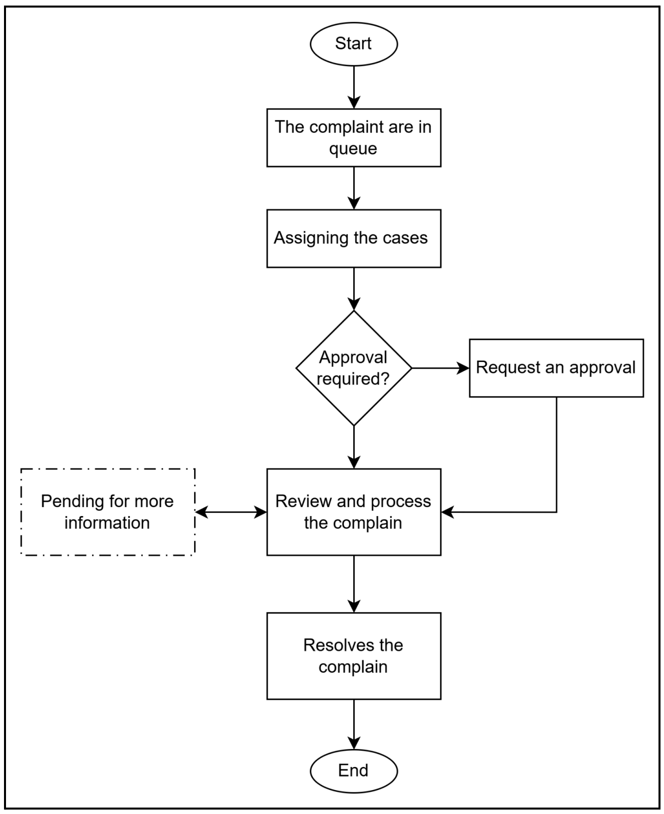

Figure 3: Admin Process Flow

The following figure illustrates the proposed administrator process flow of the system. Once a user submits a complaint or request form, the system automatically pre-routes the case to the appropriate department based on its category through rule-based classification. The administrator then receives and reviews the case, confirming or overriding the pre-assigned department before proceeding. Upon submission, the case status is initially marked as "Open" since the concern is still under review and processing.

In most cases, the assigned team handles the concern directly at their level. However, if the complaint or request exceeds the assigned office's authority or requires a higher-level decision, it is escalated to the appropriate authority, such as a department head, director, or dean, for further review and action. This reflects how authority is currently exercised at the university, where concerns are first addressed by the responsible office and elevated only when necessary. Once the assigned team begins working on the case, the administrator updates the status to "In Process" to inform the user that the concern is currently being handled.

If additional information or clarification is needed, the administrator or assigned personnel may communicate with the user through the system. In this situation, the case status will be updated to "Pending Response" until the user provides the required information. Once the complaint or request has been successfully addressed and completed, the case status will be changed to "Resolved," and the case record is retained in the system for future reference and basic administrative reporting.

1.9 Process Rules

The following rules define the operational guidelines of the proposed Smart Campus Complaint and Case Management System to ensure the proper handling, tracking, and resolution of all submitted concerns.

a. **User Registration and Access Rule:** All users must create an account before submitting a complaint or concern. Registration ensures proper identification, secure access, and reliable tracking of every submitted case.
b. **Complaint Submission Rule:** Users must complete the official complaint form and provide accurate and complete details of their concern, including the appropriate category. Supporting documents or attachments may be uploaded to strengthen the report and provide additional context.
c. **Case Creation, Initial Status, and Classification Rule:** Once a complaint is submitted, the system automatically generates a case record and assigns it an initial status of "Open." Before administrator review, the system performs a rule-based classification that scans the complaint for predefined keywords to pre-route the case to the appropriate department — for example, terms such as "aircon," "classroom," or "broken" are directed to the Facilities Office, while words like "grade," "professor," or "subject" are routed to Academic Affairs. This approach is deterministic and keyword-driven rather than machine learning-based

making it practical, transparent, and easy to audit. Since routing under the current manual process depends solely on the judgment of individual staff and is not documented anywhere, this rule standardizes and centralizes that logic. The administrator retains full authority to confirm or override the pre-assigned classification before proceeding.

d. **Assignment Rule:** The administrator assigns each case to the appropriate department or team based on its category, ensuring that every concern is handled by the personnel best suited to resolve it.

e. **Prioritization Rule:** Cases may be assigned a priority level, such as High, Medium, or Low, based on category and urgency. This addresses a gap in the current process, where concerns are handled on a first-come, first-served basis with no formal means of ranking urgency. The specific priority criteria are to be refined in consultation with the concerned offices.

f. **Escalation Rule:** Concerns are first handled at the level of the assigned office or personnel. A case is escalated to a higher authority, such as a department head, director, or dean, only when it exceeds the assigned office's authority or requires a higher-level decision. No mandatory pre-approval is required for routine cases, consistent with how authority is currently exercised at the university.

g. **Processing Rule:** Once a case is ready for action, the assigned team begins processing it, and the case status is updated to "In Process" to reflect the ongoing work.

h. **User Communication Rule:** If additional information is needed, the assigned personnel or administrator may request clarification from the user through the system. During this period, the case status is updated to "Pending Response."

i. **Response Requirement Rule:** Users are expected to respond to pending requests for information to avoid delays in resolution. The system may send reminders until the required response is provided.

j. **Resolution Rule:** Once a concern has been fully addressed and verified, the assigned personnel update the case status to "Resolved" and finalize the record.

k. **Record Retention Rule:** All resolved cases are retained in the system as permanent records. This addresses the absence of formal record-keeping in the current process and supports basic administrative reporting and the identification of recurring issues.

l. **Transparency and Monitoring Rule:** All updates, comments, and status changes are recorded in the system and made visible to the user to ensure transparency and continuous monitoring of case progress.

A. Complaint Severity and Prioritization Criteria

Table 1: Complaint Severity and Prioritization Criteria

Priority Level	Severity Description	Complaint Categories	Example Concerns	Initial Status	Expected Response Timeframe
High	Concerns that pose an immediate risk to health, safety, academic standing, or campus operations and require urgent attention from the appropriate office or escalation to a higher authority	Facilities, IT / Technical Support, Academic Affairs	Non-functioning fire exits or safety hazards; complete system or platform outage affecting ongoing classes; grade submission errors with an imminent deadline	Open → In Process	Within 24 hours of submission
Medium	Concerns that significantly affect the complainant's academic experience or access to campus services but do	Academic Affairs, Student Services, IT / Technical Support	Delayed grade encoding with no immediate deadline; unresolved enrollment concerns; malfunctioning equipment in	Open → In Process	Within 3 working days of submission

	not pose an immediate risk and can be resolved within a standard processing period		classrooms or laboratories		
Low	Concerns that are minor in nature, do not critically affect academic standing or campus operations, and can be addressed within a general administrative processing period	Facilities, Student Services	Minor facility maintenance requests such as burnt-out lighting or broken furniture; general inquiries about student services or administrative procedures	Open → In Process	Within 5 working days of submission

The Complaint Severity and Prioritization Criteria (Table 1) in the system is directly based on the administrative workflows and system lifecycle models validated by Patle et al. (2023) and Bhange et al. (2026). If complaints are typically unstructured and processed first-in-first-out, as is done in conventional university grievance structures, these systems result in considerable operational tracking delays. Patle et al. (2023) showed that structured, clearly defined categorization of incoming complaints can significantly improve the responsiveness and visibility of complaints within a department. Also, the three-tiered severity system embedded into MapúOne, namely High, Medium, and Low, for severity based on institutional safety, academic standing and operational impact has been tested and proven by Bhange et al. (2026) in centralized role based authentication and structured dashboard. These severity levels are linked to specific operating thresholds, making the whole campus grievance process more proactive, human-independent and efficient, while creating a process that is systematic, auditable and simple to manage.

B. Prioritization Assignment Rules

The following rules govern how priority levels are assigned to submitted complaints within the system.

The system initially determines priority at the time of complaint submission based on the complainant's selected category and keywords detected in the complaint description via a rule-based classification mechanism. The administrator retains full authority to review and adjust the assigned priority level before processing begins, consistent with the override authority established in Process Rule (c) and Process Rule (d) in Section 1.9.

A complaint is assigned **high** priority when its content indicates an immediate threat to health, safety, or ongoing academic activity or when the concern involves a system-wide failure affecting multiple users simultaneously. A complaint is assigned **medium** priority when it affects an individual complainant's academic progress or access to university services, but does not require emergency intervention. A complaint is assigned **low** priority when it involves a minor inconvenience, routine maintenance, or a general inquiry that can be addressed within a standard administrative processing window.

When a complaint cannot be clearly classified at submission, it is assigned Medium priority by default and flagged for administrator review before routing is confirmed. If additional information from the complainant is required to determine the appropriate priority level, the case status is updated to Pending Response, consistent with Process Rule (h) in Section 1.9, until the complainant provides the needed details.

Cases assigned high priority that remain unresolved beyond the expected response timeframe are automatically flagged on the administrative dashboard for escalation to the appropriate higher authority, consistent with the escalation pathway defined in Process Rule (f) and the Admin Process Flow in Figure 3.

These rule-based prioritization assignment rules have been directly implemented in MapúOne and reflect the algorithm and automated data pipelines validated by Jha, Karthick, et al (2023), who wrote the initial prioritization study. Jha et al. (2023) used an artificial intelligence framework to dynamically analyze incoming text string metrics and automatically produce objective priority values in a systematic manner, whereas the proposed system has a highly transparent and deterministic tracking mechanism which classifies it into a specific text string priority level. MapúOne analyzes the first unstructured text description of a submitted case and identifies predefined terms that match the categories it is programmed to look for, thus providing a systematic and rule-based approach to finding the right category for the case, rather than relying on a guess at the category by the administration. The platform's categorization pipeline closely follows the central goal of Jha et al.'s (2023) framework: to effectively automate the triage process at intake, ensure cross-departmental coordination and avoid delays in processing critical institutional tickets.

1.10 Scope And Delimitation

A. Scope

This study covers the design, development, and evaluation of a web-based Smart Campus Complaint and Case Management System (SCCCMS-MU) for Mapúa University. Specifically, it investigates how a centralized digital platform can replace the existing manual, coordinator-dependent complaint process illustrated in Figure 1 by digitizing the full complaint lifecycle—from initial submission through routing, multi-stage tracking, and final resolution—as depicted in the User Process Flow (Figure 2) and Admin Process Flow (Figure 3). The system centers on four core functions: structured complaint submission with supporting file attachments, rule-based automatic routing to the appropriate department based on complaint category and keyword classification, real-time multi-stage status tracking across the defined stages of Open, In Process, Pending Response, and Resolved, and basic administrative reporting drawn from accumulated case records.

The study investigates whether the proposed system can improve complaint submission, routing, tracking, and resolution compared to the current manual process and how end-users evaluate the system in terms of usability, functionality, and efficiency. It does not investigate the physical resolution of the underlying campus concerns themselves, as corrective action remains the responsibility of the designated university offices. The study also does not investigate the implementation of advanced intelligent features such as machine learning classification, predictive analytics, or automated decision-making, which are identified as viable directions for future development once the system has accumulated sufficient complaint data from actual university use.

The study involves two user groups directly reflected in the process flows. The first group consists of students, faculty, and staff of Mapúa University, who interact with the system as complainants through the steps shown in Figure 2 — registering an account, logging in, completing the complaint form, and monitoring their case status through the progress tracker. The second group consists of authorized university administrators, who interact with the system through the steps shown in Figure 3 — receiving queued complaints, confirming or overriding the pre-assigned department, reviewing and processing cases, updating status stages, requesting additional information where needed, escalating cases to a higher authority when required, and marking cases as resolved. Complaints involving legal disputes, disciplinary proceedings, or concerns directed at external organizations and government agencies fall outside the population of cases the system is designed to handle.

The study is conducted exclusively within Mapúa University - Makati, located at 333 Sen. Gil J. Puyat Avenue, Makati, 1200 Metro Manila, Philippines. All data gathering, system testing, pilot deployment, and evaluation activities are carried out within this campus. The system is designed for internal use within this institution and is not intended for deployment across other campuses or external organizations. The findings, while potentially informative to other Philippine higher education institutions facing similar complaint management challenges, are directly applicable only to the operational and administrative context of Mapúa University—Makati.

The study is conducted within the academic term of the 3rd term school year, 2025–2026. And the 1st term school year, 2026-2027. All phases of the study — including requirements gathering, system design, iterative development, testing, pilot deployment, and evaluation — are carried out within this period. Long-term maintenance evaluation, post-deployment performance monitoring beyond the study period, and longitudinal analysis of complaint trends over multiple academic terms are not within the scope of this study.

B. Delimitation

The system handles only campus-related complaints falling under four defined categories: facilities, academic matters, IT or technical support, and student services, consistent with the routing logic established in the process rules. It does not extend to legal disputes, formal disciplinary hearings, complaints involving external parties, or matters that fall under the jurisdiction of government agencies. The system does not perform or coordinate any physical corrective action; the resolution of the underlying concern remains entirely the responsibility of the designated university office. Advanced and optional features — including AI-based complaint categorization, machine learning severity prediction, predictive analytics, chatbot integration, voice recognition, and automated decision-making — are excluded from the current study. These are identified as future enhancements to be built upon the complaint data the system accumulates over time. Notifications are limited to in-system alerts and email only; integration with third-party communication platforms currently in use at the university, such as Microsoft Teams or Outlook, is not included.

The study evaluates the system only on usability, functionality, and efficiency as perceived by a defined set of respondents within the study period. Large-scale deployment testing, advanced cybersecurity penetration testing, and integration testing with other university systems, such as enrollment platforms, finance systems, or learning management systems, are not pursued.

The respondents are limited to a defined set of 30 students and university administrators from Mapúa University–Makati. Faculty and staff may use the system as complainants but are not included as a separate evaluated respondent group within the study's evaluation instrument. Findings are not generalized to the broader Philippine higher education population.

The study does not cover other Mapúa University campuses or any institution outside of the Makati campus. Multi-campus deployment, cross-institutional integration, and external stakeholder access are outside the scope of this investigation.

The study does not include post-deployment maintenance, long-term system performance evaluation, or multi-term complaint trend analysis. Any enhancements identified during evaluation are documented as recommendations for future work rather than implemented within the current study period.

1.11 Significance of the Study

The significance of this study lies in its contribution to improving the efficiency, transparency, and accessibility of complaint handling within Mapúa University. By developing a centralized web-based system centered on core complaint management functions — submission, routing, tracking, notifications, and reporting — the university can transition from manual and fragmented processes to an organized digital platform. This reduces the risks of lost records, delayed responses, and communication gaps, while supporting a more responsive and accountable complaint management process for students and administrators alike.

For educational institutions facing similar challenges, this study demonstrates how focused web-based solutions can strengthen communication between stakeholders and improve service delivery without requiring complex or resource-intensive features. For administrators, organized reporting and case histories allow quicker identification of recurring issues and areas requiring attention. For students, real-time tracking promotes transparency and trust in institutional processes.

For future researchers and system developers, this study serves as a practical reference for designing complaint and case management systems in academic settings. It highlights the value of prioritizing core functionality and usability over feature complexity — a foundation upon which optional enhancements such as AI categorization or predictive analytics may be built in subsequent work. Ultimately, the proposed system supports the university's goal of providing a more responsive, organized, and technology-driven environment by reducing administrative workload and improving the handling of campus-related concerns.

CHAPTER II - REVIEW OF RELATED LITERATURE

2.1 Related Studies

A thorough examination of existing literature necessitates the identification of key subject areas that directly inform and contextualize the present study. This section explores the relevant domains of knowledge drawn from published works, scholarly research, and established theoretical frameworks that collectively shape the foundation of this investigation.

A. Design of smart campus management system based on internet of things technology [1]

Li, Yuan, and Elhoseny (2021) presented a study on the design of a smart campus management system that utilizes Internet of Things (IoT) technology to enhance the efficiency, connectivity, and overall management of university campus operations. Published in the Journal of Intelligent & Fuzzy Systems: Applications in Engineering and Technology, the study responds to the increasing demand for intelligent, data-driven infrastructure in modern academic institutions and explores how IoT can serve as a foundational technology for integrating various campus systems into a unified, responsive management platform.

The authors proposed an IoT-based architecture that enables real-time data collection and monitoring across multiple campus domains, including energy management, facility monitoring, security surveillance, and environmental control. By embedding smart sensors and connected devices throughout the campus environment, the system allows administrators to gather continuous streams of operational data, enabling more informed and timely decision-making. The study highlights that the integration of IoT technology not only improves resource utilization and campus safety but also significantly reduces the manual workload placed on administrative personnel.

A central contribution of the study is its demonstration of how a centralized smart campus platform can bridge the gap between physical infrastructure and digital management systems. The authors emphasized that the success of such a platform depends heavily on seamless interoperability between devices, robust data processing capabilities, and an intuitive user interface that allows both administrators and campus users to interact with the system effectively. Furthermore, the study underscores the importance of scalability in smart campus design, ensuring that the system can accommodate the growing needs of an expanding university population.

The work of Li et al. (2021) provides a valuable theoretical and architectural foundation for the present study. While their research focuses primarily on IoT-driven facility and infrastructure management, the principles of centralization, real-time monitoring, and data-driven administration that underpin their system are equally applicable to the complaint and case management context addressed in this study. The proposed Smart Campus Complaint and Case Management System for Mapúa University draws from these principles by adopting a centralized platform approach that promotes transparency, operational efficiency, and responsive service delivery across the university's diverse administrative domains.

B. Web-based Campus Complaint Management System (WCCMS) [2]

Alifat et al. (2022) developed the Web-based Campus Complaint Management System (WCCMS), a study published in the International Journal of Computer Trends and Technology (IJCTT), which sought to address the persistent inefficiencies associated with manual complaint handling in academic institutions. The authors recognized that regardless of how well-managed a university may be, complaints from students and staff are inevitable, and that these concerns—particularly those raised by students—serve as critical indicators of institutional performance and stakeholder satisfaction.

The study argued that traditional complaint mechanisms, which often depend on physical forms, verbal communication, and informal escalation channels, are fundamentally inadequate for meeting the demands of a modern campus community. Such approaches frequently result in delayed responses, lost records, and a general reluctance among students to raise concerns due to fears of repercussion or lack of anonymity. In response, the authors proposed the WCCMS as an automated, web-based solution that enables students and staff to voice their concerns through a secure digital platform without fear of negative consequences.

The WCCMS was designed to centralize complaint submission, automate the routing of concerns to the appropriate departments, and provide users with a transparent mechanism for monitoring the progress of their submissions. The system's architecture prioritized accessibility and ease of use, ensuring that all members of the campus community — regardless of their level of technical proficiency — could interact with the platform effectively. Additionally, the study emphasized the value of complaint data as a strategic resource, noting that systematically collected grievance records can help institutions identify patterns of dissatisfaction and implement evidence-based service improvements.

The work of Alifat et al. (2022) is highly relevant to the present study, as it directly addresses the same core challenges of complaint visibility, accountability, and process efficiency within a campus setting. The WCCMS demonstrates the viability of a web-based approach to complaint management in higher education and reinforces the necessity of providing stakeholders with a transparent, accessible, and automated platform. The proposed Smart Campus Complaint and Case Management System for Mapúa University builds upon these principles while extending the scope to include real-time case tracking, inter-departmental coordination, and advanced administrative analytics suited to the institutional requirements of a Philippine technological university.

C. Beyond Complaints: A Strategic Analysis of College Grievance Redressal System [3]

The study published in the International Journal of Scientific Research in Engineering and Management (IJSREM) in June 2024 proposed a cloud-based Grievance Redressal System (GRS) designed to provide students with a structured, digital platform for submitting and monitoring academic and non-academic complaints within a university setting. The system was developed in response to the growing recognition that students in modern educational environments frequently encounter a wide range of concerns — from academic disputes to administrative difficulties — that require timely and transparent resolution mechanisms.

The GRS was built using open-source technologies, specifically PHP and MySQL, and was architected for cloud deployment to ensure accessibility, security, and ease of hosting. The platform features a user-friendly interface that simplifies the process of complaint registration and progress tracking, making it accessible to students regardless of their technical background. A key emphasis of the system is its ability to keep complainants informed throughout the resolution d/y qprocess, addressing one of the most common shortcomings of traditional grievance mechanisms — the lack of feedback and visibility for the student once a complaint has been submitted.

Beyond individual complaint resolution, the GRS incorporates reporting and analysis tools that enable university administrators to assess the nature, frequency, and distribution of grievances across the institution. These analytical capabilities support evidence-based decision-making, allowing administrators to identify recurring problem areas and implement targeted improvements to enhance the overall quality of the educational experience. The study further highlights the role of such systems in fostering institutional accountability, as they provide students with a formal channel through which the university administration can be held responsible for addressing raised concerns within a stipulated timeframe.

The findings and design principles presented in this study are directly applicable to the current research. The GRS reinforces the argument that a digitized, cloud-accessible complaint platform is essential for promoting student-centered governance and operational transparency in higher education. The proposed Smart Campus Complaint and Case Management System for Mapúa University similarly prioritizes user accessibility, real-time status tracking, and data-driven administrative reporting, while extending these capabilities to encompass a broader range of stakeholders — including faculty and staff — and incorporating inter-departmental case coordination suited to the complex organizational structure of a large technological university.

D. Development of a Web-Based Complaint Management Platform for a University Community [4]

Oguntosin et al. (2021) talked about the creation of a complaint management system that would be implemented in a web-based platform to enhance the assessment of student satisfaction and staff satisfaction in academic institutions. The study highlights the need to digitise the course of the activities on campus to eliminate manual activities that often involve complex and stressful visits to administrative offices and which lead to the lack of information tracking. Their system enables complaints to be submitted, tracked, and monitored in one centralized web application created in JavaScript and a Mongo database, allowing for efficient recording of community grievances.

The authors emphasized that a structured complaints process can enhance the effectiveness of operational responses and enhance trust in a University community. The streamlined graphical user interfaces enable students and staff members to submit concerns, as well as to access existing community complaints, thus minimizing information management problems and enhancing transparency. The study also clarifies how centralized management of complaints enables the university administrator to keep the records organized, thus allowing authorized persons to easily inspect the case files, update on the status of the resolution, and delete the unsuitable documents from the case files.

The other significant impact of the study is on the usability and modularity of the system. The system makes it easier to manage data, because users can use an intuitive community portal to indicate urgency for the administration by voting up any relevant claims. This helps resolve issues quickly and helps to reduce manual effort as much as possible from the paper tracking system. The researchers further suggested future enhancements such as integrating private messaging modules, adapting the platform for mobile devices (Android and iOS), and introducing advanced data analytics, showing the potential for university grievance platforms to evolve into highly adaptive and responsive campus tools. [9]

E. A Knowledge Management Approach towards resolving Estate Complaint Management in a higher education institute in Brunei Darussalam [5]

Azahari and Sion (2022) examined the application of a knowledge management approach in addressing estate complaint management within a higher education institution in Brunei

Darussalam. The study was published in the International Journal and responds to the growing recognition that complaint management in academic institutions must go beyond simple documentation and resolution, evolving instead into a structured process that captures institutional knowledge and drives continuous service improvement. The authors focused specifically on estate-related complaints — concerns pertaining to the physical infrastructure, facilities, and maintenance of the campus environment — and explored how knowledge management principles could be applied to streamline the handling and resolution of such concerns.

The study argued that traditional complaint handling methods commonly found in higher education institutions are largely reactive in nature, addressing concerns only after they have been formally raised without leveraging accumulated complaint data to prevent recurring issues. Azahari and Sion (2022) proposed that by integrating knowledge management strategies into the complaint resolution process, institutions can transform grievance records into valuable organizational assets. These assets, when systematically analyzed, enable administrators to identify patterns of dissatisfaction, anticipate future concerns, and implement proactive measures that enhance the overall quality of campus facilities and services.

The findings of Azahari and Sion (2022) are particularly relevant to the present study, as both works share the common objective of improving complaint management processes within an academic setting through the use of structured, technology-supported frameworks. While their research focuses on knowledge management within the context of estate and facilities concerns in Brunei Darussalam, the underlying principles of centralized documentation, pattern recognition, and evidence-based institutional improvement are directly applicable to the Smart Campus Complaint and Case Management System proposed for Mapúa University. This study builds upon these foundations by extending the scope of complaint management to encompass a broader range of campus concerns while incorporating real-time tracking, inter-departmental coordination, and data-driven analytical reporting.

F. AI-Driven Complaint Management System : An Automated Framework for Prioritizing, Tracking, and Resolving Public Service Complaints Efficiently [6]

Cholke et al. (2026) presented an AI-driven complaint management framework designed to address the persistent inefficiencies found in civic complaint management systems, particularly those arising from unstructured data, manual prioritization processes, and delayed case escalation. The study was presented at the 2026 International Conference on Emerging Smart Computing and

Informatics (ESCI) and responds to the growing demand for intelligent, automated solutions that can replace static, rule-based workflows with dynamic, data-driven complaint handling mechanisms capable of operating in real time.

The authors identified that existing digital platforms for civic engagement and complaint management continue to rely heavily on conventional, rule-based workflows that lack the capacity for dynamic severity assessment and predictive analytics. These limitations result in slow response times, unbalanced departmental workloads, and an inability to proactively address recurring or escalating concerns. To mitigate these challenges, Cholke et al. (2026) proposed a scalable AI-powered framework that integrates several advanced technologies, including natural language processing (NLP) for semantic feature extraction, a multi-task neural network for urgency and resolution time prediction, spatiotemporal clustering for duplicate complaint detection, and workload-aware dynamic routing for optimal department allocation. The framework further incorporates a hierarchical escalation mechanism that automatically elevates unresolved cases based on defined time thresholds, ensuring that critical concerns receive timely attention without requiring manual intervention.

The results of the study demonstrated substantial performance improvements across key operational metrics. The proposed system achieved an 81% reduction in average response time, a 68.8% reduction in resolution time, an improvement in workload balance index from 0.52 to 0.86, and a fake complaint detection accuracy of 92%. These findings collectively demonstrate that the integration of artificial intelligence into complaint management systems not only enhances operational efficiency but also strengthens institutional accountability, transparency, and responsiveness to public service concerns.

The work of Cholke et al. (2026) is highly relevant to the present study, as both investigations seek to improve the efficiency and responsiveness of complaint management through the use of technology-driven frameworks. While their research is situated within a civic governance context, the core principles of automated complaint classification, real-time tracking, intelligent case routing, and data-driven decision-making are directly applicable to the academic environment addressed in this study. The proposed Smart Campus Complaint and Case Management System for Mapúa University draws from these advancements by adopting a centralized, analytics-supported platform that promotes transparency, reduces resolution delays, and equips administrators with the tools necessary to manage campus concerns more effectively and proactively.

G. Decision Support Complaint Prioritization System using a Statistical Multi-Method Algorithmic approach [7]

Srivastava et al. (2025) talked about a Decision Support Complaint Prioritization System for the purpose of streamlining and standardizing complaints received by the organizations from the general public. The study highlights how the bulk and diversity of manual complaints inevitably overload the complaint system within conventional setups and result in delayed complaints being processed, and the failure to address important issues. The researchers created a prototype system that uses artificial intelligence to automatically assess and rank claims in digital form, helping to alleviate these limitations.

The authors pointed out that implementing automation onto a centralized tracking system can greatly enhance administrative visibility and response time. The system reads the incoming reports and dynamically assigns them an objective priority value using techniques such as text mining, sentiment analysis, and sentence polarity classification. The study explains how this structured data pipeline enables administrators to manage records systematically and centrally, allowing them to monitor unclosed records, queue records by priority and review high-risk records using a secure management dashboard.

A second key contribution of this study is its emphasis on ongoing decision making and operation efficiency based on data. It employs a weighted average of grievance sentiment and similarity to automatically assign complaints, thereby reducing the communications barrier between users and assigned coordinators, and reducing the amount of documentation of the manual process. The researchers also showed that the intelligent classification algorithms can be integrated into a scalable framework and that modern grievance systems can be developed into very inclusive and responsive tools for mitigating systemic risk.

H. University resources and student complaints in Malaysian higher education institutions [8]

The study by Omoola et al. (2024) examined the operational linkage between the university resource management and the handling of students' complaints in universities. This research also points out that large and complex modern universities tend to have fragmented or outdated communication systems in administration that result in frequent tracking delays. The research explores the direct impact of the availability of online and offline campus resources on stakeholder satisfaction and public trust related to an institution's grievance resolution path by using Organizational Justice Theory (OJT) as the lens through which to examine the issue.

The authors emphasized the importance of establishing organized and technologic-based complaint channels that can contribute to transparency and accountability at administrative level. The study demonstrates that student experiences are quite different, depending on the visibility and responsiveness of a university's feedback tracking, using advanced structural equation algorithms. The study clarifies the advantages of centralized records management for the administrator in recognizing systematic deficiencies and in creating accurate reports, which will reduce the ambiguity and user dissatisfaction in institutional transactions.

The other significant contribution of this study is that it centres on driving evidence-based improvements in campus governance. The system highlights the need to ensure physical infrastructure is closely linked with digital tracking tools to best enable quicker response times and reduce clerical bottlenecks. The researchers also recommended that University leadership could use summary reports of complaints as a strategic tool to use to enhance the support provided to students, and how much potential there is to evolve structured grievance metrics and make them a tool for the wider University to help them do better.

I. Effective Complaint Handling Systems, Customer Loyalty, and Sustainable Financial Performance in Pakistani Banks: A Marketing and Accounting Perspective Using PLS-SM [9]

The development of an integrated system of handling student complaints aimed at optimizing the handling of student complaints in the digital learning environment was discussed by Tshotlogo (2021). The study points out that universities are increasingly adopting more sophisticated technological systems to deliver academic subjects, while the complaint mechanisms are still mostly ineffective, waiting for a long time for a resolution and lack of fairness in the processes. To fill this gap, the researcher developed an integrated architecture to integrate complaint tracking directly into the university's core software and Learning Management System called Moodle.

The author reiterated the importance of having an open and accessible complaint tracking system to ensure institutional transparency, visibility and equity for students. The system is integrated into the digital campus portal, with clear grievance guidelines and visible monitoring paths integrated into the daily portal, so that the students can express their concerns more confidently, and eliminate the ambiguity in administration. The study also reveals how centrally managed complaints handling helps academic coordinators with clean record logs, so that student feedback is handled in a systematic manner, and none of the critical tickets are left unanswered or dropped.

A specific focus of the study has been on inclusivity and user accessibility in HE governance, which is another key contribution. The proposed framework includes the design aspects necessary to ensure the reporting tool is fair and usable for all and streamlines dialogue between citizens on campus and administrative bodies. The researcher proposed that by integrating grievance procedures into major student information systems, documentation burdens are reduced, and there is enormous potential for learning management systems to become more of an institutional accountability system.

J. Operational Excellence through an Evaluation of a Complaint Management System: A Case Study of Universiti Teknologi MARA [10]

Rahman and Mustaffar (2026) talked about the structural performance and effectiveness of governance of e-Aduan, a centralized digital complaint system developed in a large-scale university environment. The study underscores the urgent need to digitize student and staff grievance processes to remove antiquated, manual, or fragmented systems that often result in a delayed tracking process and ineffective communication. Their research assesses the possibility of a structured digital repository to enable operational excellence through systematic collection, routing, and monitoring of campus-wide concerns.

The authors emphasized the importance of a well-managed grievance system in enhancing the accountability and openness of academic institutions. The e-Aduan system also includes real-time monitoring of complaints and automation of tracking pipelines, helping stakeholders to track the advancement of their complaints, thereby relieving them of the burden of uncertainty and boosting their confidence in the responsiveness of the university. The study also outlines the concept that centralized record management creates set feedback loops enabling university leadership to have a well-structured archive, easier audit records of unresolved issues and easier application of institutional quality assurance benchmarks.

The continuous service delivery optimization and workload management aspect of the study is another contribution to be made. The system makes it easier to communicate between different campus departments by offering administrative dashboards for each department on the web, which monitor the efficiency of resolution and performance indicators. This system, which is data-driven, alleviates the physical effort required to manually document, and identifies areas where problems are likely to appear to allow a more targeted approach. The researchers also showed how grievance tools can be developed into proactive tools for long-term organization excellence through systematic review processes.

K. An integrated student complaints handling framework: A learning management system for an Open University in Botswana [11]

Under this study, Khan et al. (2025) delved into the emergence and critical success factors of an effective complaint handling system in an institutional service environment. Although the study is focused on the financial services industry, there are universal issues with the operation of traditional or poorly managed grievance processes that drive stakeholders away, such as lack of transparency, friction in the process, and lack of tracking channels. With the method of variance-based structural equation modeling, the study is able to extract and validate nine foundational determinants that are required to build an optimized, reliable complaint system architecture.

The authors pointed out that the key components that affect stakeholder confidence and institutional loyalty are visibility and accessibility of the platform. They found that an effective grievance solution should offer a clearly visible and easy-to-use filing system, maintaining the user with automated solutions and real-time updates. By showing how central data management can cut down tracking time and deal with disagreements fairly, the study can minimize any delays and communication barriers in handling grievances that often occur with informal or legacy grievance processes.

The study's second key contribution is its emphasis on the use of organized complaint records as a strategic resource to enable continual improvement. It affirms that systematic audit logs and audit log analysis of the frequency and nature of concerns provided to management can help them to improve their services by carrying out evidence-based service recovery processes and preventing or reducing systemic operational failures. The researchers also found that the use of responsive tracking metrics significantly reduces users' frustration to the point that the principles of transparent and centralized complaint resolution are essential to the successful operation of any large-scale service network and to the satisfaction of its users.

L. Semantic-Based Analysis in Complaint Management System with Analytics [12]

Alterado (n.d.) developed a Semantic-Based Analysis in Complaint Management System with Analytics at Palawan Polytechnic College Inc. (PPCI) in the Philippines, responding to the inefficiencies of the institution's existing manual grievance process. In the existing system, students were required to physically visit the guidance office and submit complaints through written statements or drop boxes, a process that was time-consuming, offered no complaint acknowledgment, and provided no mechanism for tracking resolution progress. The absence of a structured complaint channel resulted in poor record management and limited accountability between students and administrators.

To address these gaps, the researcher developed a two-component system consisting of a mobile application for student complainants and a web-based platform for guidance counselors, department heads, quality management resources, and IT administrators. The system employed three core algorithms: text mining to detect and filter complaints containing foul language, semantic analysis to automatically determine which department a complaint should be routed to, and visual analytics to generate comparative reports on complaint frequency by department and time period. A notification module was also integrated to alert concerned personnel when a complaint remained unresolved beyond 24 hours.

The system was tested by 100 randomly selected students across 10 departments, and all modules were confirmed to be fully functional. Evaluation conducted using the ISO 9126 standard assessed the system in terms of usability, user-friendliness, reliability, portability, and accuracy, with respondents rating the system 4.7 out of 5, described as Excellent. The study demonstrates that semantic-based text processing and automated complaint routing are viable and effective approaches for digitizing grievance management within a Philippine higher education institution.

M. Development of an Online Crime Management & Reporting System [13]

Ganiron Jr. et al. (2019) developed an online crime management and reporting system for Barangay Bangkal in Makati City, Philippines, responding to the widespread problem of unreported crimes caused by the inconvenience of physically traveling to police stations, fear of personal involvement, and limited public awareness of crimes occurring within the community. The study recognized that distance, accessibility, and lack of secure reporting channels were primary factors discouraging citizens from filing complaints through traditional manual methods, which were also time-consuming and prone to delays in action.

The researchers developed a web-based platform that allowed registered users to submit crime reports and report missing persons online, while also enabling offline reporting through mobile phones via GSM technology. The system featured role-based access controls, where registered users could submit reports, view announcements, and communicate with administrators through a built-in chat box, while administrators managed submitted reports, posted public announcements, and monitored crime activity. A summary chart was also incorporated to help administrators identify frequently occurring crimes within a given period. The system was built using PHP, Bootstrap, and MySQL, and followed the Modified Waterfall Methodology across phases of brainstorming, requirements analysis, system design, implementation, deployment, and maintenance.

The system was evaluated by 50 respondents consisting of 15 barangay officials and 35 residents using the ISO/IEC 25010:2011 standard across five criteria: functionality, efficiency, usability, reliability, and security. Both groups rated the system "Strongly Satisfied" across all criteria, with barangay officials and residents recording overall means of 4.75 and 4.69 for functionality, 4.58 and 4.70 for efficiency, 4.42 and 4.65 for usability, 4.78 and 4.68 for reliability, and 4.55 and 4.65 for security respectively. Statistical testing further confirmed no significant differences between the two groups' assessments, indicating consistent satisfaction across user types. The study demonstrates that centralized, web-based reporting platforms with role-based access and secure digital submission channels are effective in improving transparency, accessibility, and responsiveness in complaint and incident management within a Philippine community context.

N. Technology Management Framework for Smart University System in the Philippines [14]

Fernando-Raguro et al. (2021) developed a technology management framework for smart university systems in the Philippines, conducted across selected universities in Metro Manila. The study was motivated by the growing recognition that smart city principles could be applied to university operations, transforming traditional academic institutions into more efficient, responsive, and technology-driven environments. Using a mixed-methods approach combining the Nominal Group Technique for qualitative data and survey questionnaires for quantitative data, the researchers gathered responses from 160 participants consisting of students, teachers, university administrators, and IT experts to assess the smartness levels of participating institutions.

The resulting framework is organized around three pillars—Technology Infrastructure, Educational Pedagogy, and Government Collaboration—and five major criteria: Learning Environment, Educational Technology, Academic System, Governance, and Health Services. Under the Governance criterion, the framework specifically identifies the need for administrative services management systems, financial management tools, and university-wide risk management platforms as essential components of a smart university. Assessment results revealed that most Metro Manila universities were still at the Defined Level, meaning they were in the early stages of formulating strategic plans for smart university adoption, and that no existing accreditation instrument in the Philippines formally recognizes or evaluates smart university status—identifying this as a significant gap in the country's higher education quality assurance landscape.

O. Help Desk Management System [15]

Help desk systems have long been recognized as essential infrastructure in organizational settings. Hu (2013) described help desk software as a solution application used for managing an organization's support needs, accessible to customer support personnel who can direct requests to appropriate servicing departments. Middleton and Marcella (1997) further characterized help desk automation as one of the earliest applications of knowledge-based systems, distinguishing it from simpler record-keeping tools by framing it as a knowledge distribution mechanism with strategic implications. Despite this recognition, Kane (2001) noted that manual workflows remain costly and inefficient in academic environments, and that automated systems, while theoretically available, were not yet widely adopted due to unresolved concerns about their real-world performance. Sheehan (2007) reinforced this by observing that users tend to form their broader perception of an entire IT organization based on their help desk experience, making responsiveness and quality not merely operational concerns but reputational ones.

Research on more advanced implementations further exposes the limitations of existing approaches. Byeong et al. (2008) demonstrated that AI-enhanced help desk interfaces could improve response accuracy, but acknowledged that such systems require continuous updating of their underlying knowledge base to remain effective. A separate framework studied for service fault diagnosis aimed to reduce resolution times and support compliance with Service Level Agreements, yet its scope remained narrowly focused on fault management rather than delivering an integrated support experience for end users. Similarly, the SysRef web-based tool discussed by one research group showed the value of building on existing infrastructure, but was limited to a single institutional context with little evidence of broader applicability. Symantec Corporation (2009) argued that true service desk effectiveness requires automating manual steps, enforcing best practices consistently, and enabling user self-service — capabilities that many existing implementations still fall short of delivering in full.

Taken together, the existing literature reveals that while the benefits of automated help desk systems are well established in theory, critical gaps persist in practice — particularly in academic and institutional settings. Existing systems tend to address either the technical fault management side or the user-facing service side in isolation, rarely integrating both into a cohesive solution. The absence of a unified, automated system capable of extending service availability, reducing direct staff dependency, and providing measurable performance outcomes remains a consistent finding across the reviewed works, underscoring the need for a more comprehensive approach to help desk management in educational environments.

2.2 Related System

A system is broadly defined as a set of interrelated components working together toward a common goal or function. According to Laudon and Laudon (2020), systems within organizational contexts are structured to process inputs, manage operations, and produce outputs that support decision-making and operational efficiency. Understanding the nature and behavior of systems is fundamental to analyzing how technological and procedural frameworks are designed, implemented, and evaluated in various settings.

A. Student Review and Complaint System [16]

Goyal et al. (2025) discussed how it is developing a Student Review and Complaint System designed to bridge the communication gap between students and the governance bodies in the higher education sector. The study highlights the need to digitize the students' feedback systems, making them more effective as they overcome low tracking efficiency and lack of accountability in the manual or informal feedback mechanism. They have an online platform that offers a structured route through which complaints in relation to the behaviour of faculty members, the condition of the facilities on campus, or issues related to grades can be made directly with the institution, so that matters can be addressed in an orderly manner and not forgotten

The authors emphasized that an open and automated review system can significantly enhance the responsiveness of institutions and re-establish trust with students. Students can access the actual operation progress of their cases on real-time tracking dashboards, which reduces administrative confusion and the need to keep going back to the offices to keep track of their cases. The study further elucidates how the centralization of record management enables the university administrators (Heads of Departments (HODs) and principal committees) to keep well-organized digital logs, which helps in assessing the performance of the departments, flagging any repeating structural issues, and analyzing past cases.

One of the significant findings of the study is that it focuses on continuous quality improvement and system accessibility. The application uses state-of-the-art front-end and back-end technology to streamline interactions between citizens of the campus and the campus decision makers, thereby reducing the amount of documentation normally required for a manual grievance trail. The researchers went on to state that the incorporation of future capabilities, including automated machine learning sentiment analysis, predictive campus trend analytics, and cross-department notification routing, demonstrates the powerful potential to turn review platforms into full-fledged institutional excellence tools. [10]

B. Grievance Management System: Streamlining Complaint Resolution [17]

Priyanka and Mahesh (2025) discussed the development of a grievance management system intended to improve how universities manage student complaints and concerns. The study emphasizes the importance of digitalizing grievance procedures to replace traditional manual processes that often cause delays, lost records, and poor communication between students and school offices. Their system allows complaints to be submitted, tracked, and monitored through a centralized platform, ensuring that each concern is directed to the appropriate department or authority for proper action.

The authors highlighted that an organized grievance platform can improve transparency and accountability within academic institutions. Through real-time complaint tracking, students are able to monitor the progress and status of their submitted concerns, reducing uncertainty and increasing trust in the institution's response process. The study also explains that centralized complaint management helps administrators maintain organized records, making it easier to review cases, monitor unresolved issues, and generate reports when needed.

Another important contribution of the study is its focus on accessibility and efficiency. The system simplifies communication between students and administrators by providing a structured method for submitting and resolving complaints. This supports faster response times and reduces the workload associated with manual documentation. The researchers further suggested future enhancements such as AI-based complaint categorization, multilingual support, and mobile accessibility, showing the potential for grievance systems to evolve into more intelligent and inclusive campus management tools. [11]

C. Efficient Campus Solutions: A Journal on Enhancing College Complaint Management [18]

Patle et al. (2023) presented a campus complaint management system designed to improve how complaints are submitted, assigned, tracked, and resolved within a college setting. The study is centered on a web-based platform that gives both administrators and complainants clear interfaces, secure authentication, detailed complaint forms, real-time status tracking, and notification support, so the complaint process becomes more organized and visible to users.

The paper also emphasizes the administrative side of complaint handling. Its dashboard allows officials to monitor complaint trends, track resolution times, and assess departmental performance, which means the system is not only a reporting tool but also a way to evaluate institutional responsiveness. The authors report that the platform is intended to improve transparency, accountability, and overall satisfaction in the college community, and they describe continuous improvement and feedback mechanisms to adapt the system to changing campus needs. [12]

D. A Survey on Student Grievance Redressal System [19]

Meshram et al. (2023) discussed a student grievance redressal system designed to provide a structured and digital approach for handling student concerns within academic institutions. The study explains that many students struggle to express complaints effectively through traditional or informal methods, which may lead to unresolved issues and dissatisfaction. To address this problem, the researchers developed a system that allows students to submit grievances, monitor complaint progress, and communicate concerns through a centralized platform. The paper emphasizes the importance of creating an organized reporting channel that improves accessibility and encourages students to voice their concerns more confidently.

The study also highlights the role of reporting and analytical tools in improving institutional responsiveness. According to the authors, the system not only manages complaint submissions but also helps administrators analyze complaint patterns and recurring issues within the university. Through timely complaint handling and continuous progress updates, students remain informed while administrators gain useful data for improving services and policies. This study is relevant because it demonstrates that grievance management systems should include monitoring, tracking, and reporting features to support efficient complaint resolution and help institutions identify long-term improvements for the educational environment. [13]

E. Student Complaint Management System [20]

Al-Waeli and Hassan (2022) talked about the creation of a Student Complaint Management System that would revolutionize the way higher education institutions collect, monitor, and address structural complaints. The study points out that technology has changed the way people have cooperated and communicated with one another in the world, but traditional university procedures on complaints and the management of facilities or academic issues remain paper-based. This old processing model constantly leaves all of the administrative systems of the institution vulnerable to tracking delays, data loss, and operating inefficiencies between students and administration.

The authors highlighted the importance of developing an automated grievance framework as a key element of structural reliability, transparency, and internal accountability in a university network. The software proposed would be a structured web-based system that would allow a direct collection pipeline for registering individual student complaints, mapping them to clear service-measure metrics, and dynamically tracking them. The study outlines the positive impact of centralized progress tracking for university administrators: having structured, uncompromised audit histories to easily review cases in the future, avoid missing critical student tickets, and quickly generate reports for upper-level decision-makers in higher education.

The study has also given a specific focus on enhancing the speed of operational execution and optimizing the management of campus resources. The application's user-friendly, structured design facilitates communication between student representatives and facility managers and can help to reduce the administrative workload associated with manual campus documentation. Moreover, the researchers described the system's scalability over time, explaining how contemporary campus systems can adapt over time to accommodate new feature sets to keep up with the changing needs of the academic community. [8]

F. CampusFix—Smart Campus Complaint Management System Using Machine Learning [21]

Reddy et al. (2026) developed CampusFix, a smart campus complaint management system that leverages machine learning to automate and streamline the handling of complaints within a university setting. The study was presented at the 2026 Second International Conference on Multi-Agent Systems for Collaborative Intelligence (ICMSCI) held in Erode, India, and addresses the growing need for intelligent, technology-driven solutions to replace inefficient manual complaint processes in academic institutions.

The primary objective of CampusFix is to provide a unified platform through which campus stakeholders can submit complaints that are automatically categorized and routed to the appropriate department using machine learning algorithms. By eliminating the need for manual complaint classification, the system significantly reduces response times and minimizes the risk of misdirected or unattended concerns. The integration of machine learning further enables the system to recognize patterns in submitted complaints, allowing administrators to proactively identify recurring issues and implement targeted interventions.

The proposed system incorporates key features such as automated complaint classification, real-time status tracking, and a centralized administrative dashboard for case monitoring and resolution management. These functionalities collectively address common pain points in traditional complaint handling, including delayed responses, lack of transparency, and fragmented documentation. The authors demonstrated that the application of machine learning in complaint management not only improves operational efficiency but also enhances the overall user experience by providing timely feedback and updates to complainants.

The findings of Reddy et al. (2026) are directly relevant to the present study, as both works share the common goal of digitizing and optimizing complaint management within a campus environment. While CampusFix emphasizes the role of machine learning in automating complaint classification, the proposed Smart Campus Complaint and Case Management System for Mapúa University expands on this foundation by incorporating inter-departmental coordination, supporting documentation submission, and data-driven analytical reporting tailored to the specific operational context of a Philippine higher education institution.

G. RailNivaran+: A Efficient Complaint Management System [22]

Patil et al. (2025) presented RailNivaran+, an efficient complaint management system developed and introduced at the 2025 International Conference on Future Technologies (ICFT). The study addresses the operational challenges commonly associated with large-scale public service complaint management, particularly within the context of railway systems where the volume, diversity, and urgency of passenger complaints demand a structured, technology-driven approach to ensure timely resolution and sustained service quality. The authors recognized that conventional complaint handling mechanisms in public service sectors are frequently characterized by manual processing, fragmented communication channels, and inadequate tracking systems that collectively contribute to delayed responses and diminished user satisfaction.

In response to these challenges, RailNivaran+ was designed as an efficient and scalable complaint management platform that centralizes the submission, categorization, routing, and resolution of complaints within a unified digital framework. The system emphasizes the importance of structured complaint workflows that reduce administrative bottlenecks, improve inter-departmental coordination, and ensure that each concern is directed to the appropriate authority for prompt action. By digitizing the entire complaint lifecycle, the platform aims to enhance transparency for complainants while simultaneously equipping administrators with the tools necessary to monitor case progress, manage workloads, and generate actionable insights from accumulated complaint data.

The work of Patil et al. (2025) is relevant to the present study as it reinforces the universal applicability of centralized, digital complaint management frameworks across varying institutional and sectoral contexts. While RailNivaran+ operates within a public transportation environment, the foundational principles of streamlined complaint submission, real-time status tracking, structured case routing, and data-driven administrative oversight align closely with the objectives of the proposed Smart Campus Complaint and Case Management System for Mapúa University. This study draws from these principles in designing a complaint management platform tailored to the academic environment, where diverse stakeholder concerns spanning facilities, academic affairs, technical support, and student services require an equally structured and responsive management approach.

H. Smart Complaint Management System [23]

Bhange et al. (2026) proposed a smart complaint management system for the optimization of public concerns registration, allocation, and resolution in the institutional environment. The study also illustrates the key challenges with the traditional grievance process: a reliance on verbal reporting, inconsistent communication channels, and manual tracking, leading to significant administrative delays and loss of stakeholder trust. To address these operational deficiencies, the researchers developed a centralized web-based platform to digitalize the entire stage of the complaint to make it transparent, structured, and monitored in real time.

The authors pointed out that the creation of transparent and specific interfaces for citizens and officials significantly enhances transparency and operational efficiencies of institutions. Accessible to users via a secure login and role-based authentication system, elaborate complaint forms allow users to add supporting information and track the progress of their complaints at different stages, such as Pending, In Progress, and Resolved. According to the study, this centralized approach enables administrators to keep digital logs clean and allows users to quickly search logs, assign tasks to various departments, and mark tickets as resolved.

A further merit of the study is that it has been based on the use of complaint information as an administrative instrument for performance monitoring and assessment. Reporting engines and dashboard statistics are built into the system, enabling management to track the trends of grievances, accurate resolution times and departmental productivity. This is a reliable way that reduces the risk of human mistakes that can occur using paper tracking and maintains ongoing communication through automated notifications. The researchers also highlighted the highly adaptive nature of the modular design, describing how digital complaint networks can be easily integrated with future tools and techniques of analysis to ensure accountability for institutions over time.

I. Machine Learning based Student Complaint and Priority Determination Support System [24]

In higher education, providing support for students is a critical task, and Oliver et al. (2025) introduced an "intelligent Student Complaint and Priority Determination Support System". While digital learning environments are growing in number, many universities still use the traditional paper-based grievance process.Although digital learning environments are proliferating, many universities are still stuck with the traditional paper-based grievance process. This antiquated model leaves institutional networks continually vulnerable to delayed tracking, lost logs and ineffective communications between students and campus administrators.

The authors highlighted the fact that the incorporation of machine learning into the campus workflow is a key part in building structural reliability, transparency, and internal accountability. The system uses the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework and applies data mining and artificial intelligence to analyze student queries, automatically prioritizing the queries according to the urgency in the text. The research goes on to clarify that the centralised progress tracking in conjunction with an automated responder module not only results in a secure tracking history for the administrative teams but also enables them to swiftly generate reports and not miss out on critical tickets.

The other key feature of the study is its adherence to the principles of object-oriented programming to ensure the system's scalability and modularity. The web application makes communication more convenient thanks to a structured knowledge base, which can process common questions instantly, thus relieving the administrative burden on the university staff. The researchers also proposed the possibility of its easy adoption of progressive functionality over time, showing how the campus complaint network could grow to the needs of an expanding academic body.

J. Web-Based Integrated Student Complaint System [25]

Omoola et al. (2024) explored the operational relationship between university resource management and student complaint handling within higher education institutions. The study highlights that as contemporary universities grow in size and structural complexity, administrative offices often rely on fractured or legacy communication procedures that cause severe tracking delays. Using Organizational Justice Theory (OJT) as a guiding framework, the research examines how the availability of both online and offline campus resources directly dictates stakeholder satisfaction and public trust in an institution's grievance resolution path.

The authors highlighted that establishing organized, tech-driven complaint channels plays a vital role in maintaining transparency and accountability across administrative departments. By analyzing data through advanced structural equation algorithms, the study shows that students' experiences vary significantly based on how visible and responsive a university's feedback tracking is. The research explains that centralized records management helps administrators identify systematic deficiencies and compile accurate reports, which minimizes ambiguity and reduces user dissatisfaction during institutional transactions.

Another important contribution of the study is its focus on driving evidence-based improvements in campus governance. The system underscores that physical infrastructure must be tightly integrated with digital tracking tools to fully support faster response times and lessen clerical bottlenecks. The researchers further suggested that university leadership should treat complaint summaries as a strategic resource to optimize student support services, showing the immense potential for structured grievance metrics to evolve into tools for broader institutional excellence.

2.3 Summary of Related Literature

The literature surveyed for the proposed system focuses on the migration of institutional grievance processes from legacy processes to digitized processes based on data. Paper-based and manual tracking systems are routinely found to cause significant administrative backups, data loss and delayed resolution time, ultimately leading to loss of stakeholder trust. Modern businesses can ensure long-term record integrity, reduce the burden on clerical tasks, and regain stakeholder trust by using web-based or cloud-based solutions, providing full transparency of processes and real-time tracking portals.

One of the key findings in the international literature is the shift towards proactive institutional governance as opposed to reactive complaints filing. It has been proven again and again that structured databases and automated classification engines are not just loggers – these tools transform passive complaint data to operational assets. Advanced frameworks involve the use of digital tracking dashboards, digital rule-based text classification and machine learning logic to enable automated domain routing, departmental workload balancing and automated case escalation. Such technological functions enable administrative leaders to appraise departmental turnaround measurement, anticipate with certainty seasonally induced system congestion and to make a systematic audit of structural anomalies prior to their development into broader system failures.

Importantly, the newly introduced literature locally confirms the high usability, functional efficiency and performance viability of the centralized digital platforms in the institutional setting in the Philippines. The feasibility of semantic-based text processing, the use of secure role-based access tokens, and electronic submission mechanisms have been demonstrated in a number of regional studies across local education environments and community settings to address the accessibility "catch and the delay" issues associated with a traditional dropping system. Compared with the industry standards for software, these 'localized' architectures score high satisfaction levels for both end-users and administrators, indicating that centralized Web repositories are a sound source of evidence-based, transparent administrative response.

2.4 Synthesis of the Related Literature

The literature reviewed indicates a clear academic consensus on the need to change the way campus grievance handling is done from the traditional manual processes to digital, centralized processes. Manual tracking methods that require paper, verbal reporting, drop boxes and disjointed communication pathways are always fraught with serious tracking delays, data loss due to clerical work, and a lack of accountability during the operation that inevitably weakens stakeholder trust. Such changes to workflows ensure long-term data integrity, decrease the burden of time-consuming administrative processing, and help rebuild trust in a community by providing absolute transparency with real-time tracking interfaces. But by analyzing the current state of software models, it becomes clear that there are important levels of abstraction between general web-based applications, sophisticated algorithmic applications and local administration requirements for technological institutions.

The technical characteristics of the reviewed literature can be contrasted on four dimensions to clearly put into context the functional capabilities offered by previous works and the ones introduced by this proposed system:

- **Complaint Tracking and Severity Scoring:** Common university platforms, such as Alifat et al. (2022), Goyal et al. (2025), Oguntosin et al. (2021), Rahman and Mustaffar (2026), and Tshotlego

(2021), capture and record information in cases in a defined lifecycle, such as Pending, In progress, or Resolved. These systems, however, do not have any objective prioritisation rules and it is up to university staff to manually review unstructured text logs to decide on the urgency of a case. The Smart Campus Complaint & Case Management System on the other hand will integrate the tracking of the lifecycle stages with a structured severity classification system at the very intake stage, and dynamically calculate the priority of the complaint according to the defined institutional parameters.

- **Manual vs. Automated Routing:** Traditional approach to dealing with campus applications, such as the one used by Bhange et al. (2026), Al-Waeli and Hassan (2022) and Khan et al. (2025), are limited by manual routing processes that involve manual assessment and routing of all incoming tickets by primary administrators. On the other hand, more sophisticated civic or predictive types, such as CampusFix (Reddy et al., 2026) and the platform for decision support by Srivastava et al. (2025), are primarily used as triage tools on the front end that do not have multi-stage tracking pipelines. The proposed solution aims to address this issue by creating a clear and rule-based parsing system which routes tickets to specific campus domains and also provides audit trails.
- **Basic Reports vs. Predictive Analytics:** Current grievance systems, like the ones analyzed by Meshram et al. (2023) and Patle et al. (2023), limit their administrative reporting components to simple, more summative reports on past complaint volumes and turnaround times. The proposed platform extends this descriptive base with predictive analytics and ongoing feedback rules on the administrative dashboard using case history stored in the platform to predict possible congestion in the system at any given time across the campus and recurring structural trends.
- **User Demographics and System Scope:** Most of the academic grievance systems in the literature are student-focused, and either do not include any representation from the university staff, or direct employee concerns to an uncoordinated general queue. The system being proposed expands these demographic boundaries by bringing together all students, faculty and staff members in a single, coordinated digital environment. In addition, deliverability of the project is guaranteed by a clear scope of work that explicitly excludes external legal issues, formal disciplinary proceedings and direct connectivity with complex enrollment and financial system.

The proposed system combines these four technical aspects to a coherent system and thus directly addresses the structural fragmentation, which was established in the previous research. The multi-role accessibility, combined with the ability to collect evidence through multimedia and the automation of triage logic and the prediction of tracking analytics, creates an institutionally aligned platform. In particular, the Continuous Evaluation & Feedback Loop means that the completed historical summaries can be seamlessly integrated back into the core platform database, which will enable a more adaptive, optimized, and responsive case management system that is exactly suited to the organizational structure of Mapúa University.

2.5 Gap Analysis

The following table presents a synthesis of the reviewed literature and related systems, comparing each study against key features relevant to the proposed Smart Campus Complaint and Case Management System for Mapúa University (SCCCMS-MU). The comparison focuses on seven dimensions: digital complaint submission and tracking, severity scoring, automated routing, cross-departmental coordination, stakeholder scope, and institutional context. As shown, most existing systems address only a subset of these features, predominantly serving students alone while lacking automated routing, cross-departmental workflows, and adaptation to a Philippine higher education context. The proposed SCCCMS-MU directly addresses these gaps by integrating the features identified as absent or incomplete across the reviewed works into a single, institutionally aligned platform.

Table 1: Synthesis and Gap Analysis of Related Literature and Systems

Comparison of key features across reviewed studies and the proposed Smart Campus Complaint & Case Management System for Mapúa University (SCCCMS-MU)

Legend: **(✓)** Present **(~)** Partial or future work **(✗)** Absent

Study	Gap Identified	Complaint Management		Routing		Stakeholder Scope		Con text
		Digital Submission & Tracking	Severity Scoring	Automated Routing	Cross-Department Coordination	Students	Faculty & Staff	PH HEI
RELATED LITERATURE								
Li et al. 2021 | IoT smart campus	No complaint intake or case tracking; facility-only scope	✗	✗	✗	✗	✗	✗	✗

Alifat et al. 2022 | WCCMS	Students only; no severity scoring or predictive analytics	✓	✗	✓	✗	✓	✗	✗
IJSREM 2024 | Cloud-based GRS	Students only; no automated routing or cross-dept. coordination	✓	✗	✗	✗	✓	✗	✗
Oguntosin et al. 2021 | Web complaint platform	Manual prioritization via community voting; no routing logic	✓	✗	✗	✗	✓	✗	✗
Azahari & Sion 2022 | Estate complaints, Brunei	Facility complaints only; no real-time tracking or cross-dept workflow	~	✗	✗	✗	~	✗	✗
Cholke et al. 2026 | AI-driven civic CMS	Civic sector only; not adapted to academic workflows or HEI structures	✓	✓	✓	~	✗	✗	✗
Srivastava et al. 2025 | Decision support prioritization	No case routing, no complainant dashboard, general org context	✓	✓	~	✗	✓	✗	✗
Omoola et al. 2024 | University resources &	Theoretical only; no working system;	~	✗	✗	✗	✓	✗	✗

complaints, MY	students only							
Tshotlego 2021 | LMS-integrated complaints, Botswana	LMS-dependent; students only; no severity classification or routing	✓	✗	✗	✗	✓	✗	✗
Rahman & Mustaffar 2026 | e-Aduan, UiTM	No severity scoring; manual routing; no predictive analytics	✓	✗	~	~	✓	~	✗
Khan et al. 2025 | Complaint handling, banking	Theoretical framework only; non-academic, non-campus sector	~	✗	✗	✗	✗	✗	✗
RELATED SYSTEMS								
Goyal et al. 2025 | Student review & complaint system	Students only; no automated routing; AI deferred to future work	✓	✗	✗	✗	✓	✗	✗
Priyanka & Mahesh 2025 | Grievance management system	Students only; no severity scoring; AI and mobile deferred	✓	✗	✗	✗	✓	✗	✗
Patle et al. 2023 | Campus complaint management	No automated routing; no severity scoring; no cross-dept workflow	✓	✗	✗	✗	✓	✗	✗

Meshram et al. 2023 | Survey on grievance redressal	Students only; descriptive analytics only; no routing intelligence	✓	✗	✗	✗	✓	✗	✗
Al-Waeli & Hassan 2022 | Student complaint mgmt. system	Students only; no severity scoring; no cross-dept routing	✓	✗	✗	✗	✓	✗	✗
Reddy et al. 2026 | CampusFix — ML campus CMS	Students only; no cross-dept coordination; no predictive analytics	✓	~	✓	✗	✓	✗	✗
Patil et al. 2025 | RailNivaran+ — railway CMS	Public transport sector only; no academic workflows or HEI roles	✓	✗	✓	✓	✗	✗	✗
Bhange et al. 2026 | Smart complaint mgmt. system	General institutional scope; no severity scoring or predictive analytics	✓	✗	✗	✗	✓	✗	✗
Oliver et al. 2025 | ML student complaint priority system	Students only; no cross-dept routing or institutional analytics	✓	✓	~	✗	✓	✗	✗
LOCAL STUDIES								
Alterado n.d. |	Students only; no	✓	~	✓	✗	✓	✗	✓

Semantic-based CMS, PPCI Philippines	cross-dept coordination; no severity scoring beyond routing; single institution scope							
Ganiron Jr. et al. 2019 | Online Crime Reporting System, Makati	Community/ barangay context only; no academic workflows; no severity scoring or cross-dept case management	✓	✗	✗	✗	✗	✗	✓
Fernando-Raguro et al. 2021 | Tech Management Framework, Metro Manila HEIs	Framework level only; no working complaint system; no tracking. routing, or case resolution features	✗	✗	✗	✗	~	~	✓
Masongsong & Damian 2016 | Help Desk Management System, Metro Manila	IT support focus only; no severity scoring or cross-dept coordination; limited to technical concerns	✓	✗	✗	✗	~	~	✓
PROPOSED SYSTEM								
SCCCMS-MU Mapúa University	Addresses all identified gaps within a PH HEI context	✓	~	✓	✓	✓	✓	✓

2.6 Conceptual Framework

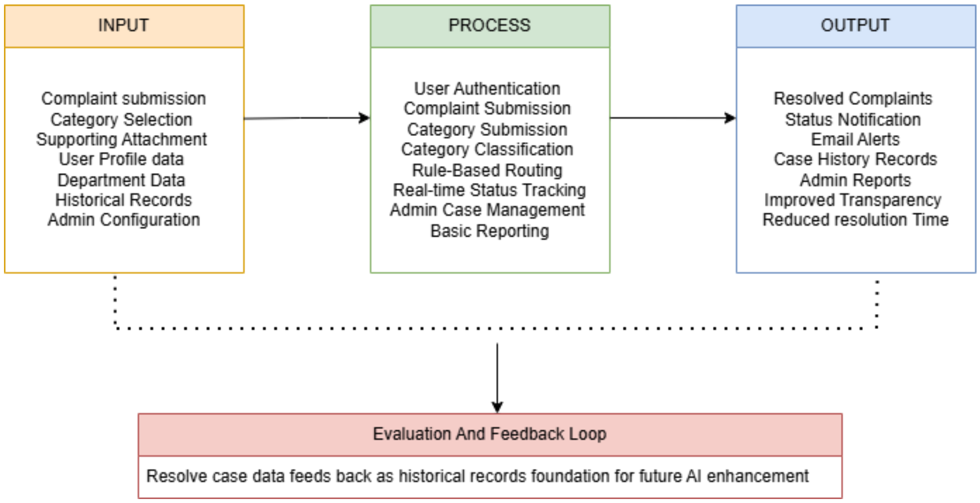

Figure 4: Conceptual Framework

The conceptual framework of the proposed Smart Campus Complaint and Case Management System for Mapúa University (SCCCMS-MU) follows an Input-Process-Output (IPO) model supported by an Evaluation and Feedback Loop.

Input consists of all data and information entered into the system by its users and administrators. This includes complaint submissions from students, faculty, and staff; category selections; supporting file attachments such as images and documents; user profile and account information; department configurations; and administrator settings used to manage routing rules and case assignments.

Process represents the core operations performed by the system upon receiving user input. These include user authentication and access validation, complaint submission and category-based classification, rule-based routing that automatically directs complaints to the appropriate department based on complaint category, real-time status tracking across defined case stages of Open, In Process, Pending Response, and Resolved, administrative case management through a centralized dashboard, and basic reporting that summarizes complaint volume, resolution times, and departmental case activity.

Output encompasses the results produced by the system for both complainants and administrators. These include resolved complaints with documented case histories, real-time status notifications delivered through in-system alerts and email, an organized complaint record repository for administrative reference, and basic analytical reports that support operational monitoring and decision-making.

The **Evaluation and Feedback Loop** ensures that resolved complaint data and user feedback are fed back into the system as historical records. These records support ongoing administrative reporting and serve as a foundation for future data-driven enhancements, including machine learning classification and predictive analytics, which are identified as viable directions for subsequent development once sufficient complaint data has been accumulated.

2.7 User Matrix

Table 2: User Matrix

FEATURES	USERS	ADMINISTRATION
ACCOUNT ACCESS		
Register Account	☑	☑
Login/Logout	☑	☑
Manage Profile	☑	☑
CASE		
Submit Complaint/Concern	☑	☑
View Submitted Complaints	☑	☑
Track Complaint Status	☑	☑
Receive Notifications/Updates	☑	☑
View Complaint History	☑	☑
Search Complaints	☑ **(Own Complaints Only)**	☑
Delete Complaint		☑
View All Complaints		☑
Assign Cases to Personnel		☑
Update Complaint Status		☑
Respond to Complaints	☑	☑

Manage Complaint Categories		☑
Generate Reports and Analytics		☑
View Dashboard Statistics		☑
Manage User Accounts		☑
Archive Closed Cases		☑
Export Reports		☑

CHAPTER III - METHODOLOGY

This chapter presents the methodological framework employed in the design, development, and evaluation of the Smart Campus Complaint and Case Management System for Mapúa University. It outlines the software development methodology and model adopted by the research team, describes the phases through which the system will be built and deployed, and details the research design, locale, respondents, instruments, and data gathering procedures used to assess the system's effectiveness. The approaches described in this chapter were carefully selected to ensure that the development process remains systematic, stakeholder-driven, and aligned with the academic and operational objectives of the study.

3.1 Software Development Methodology

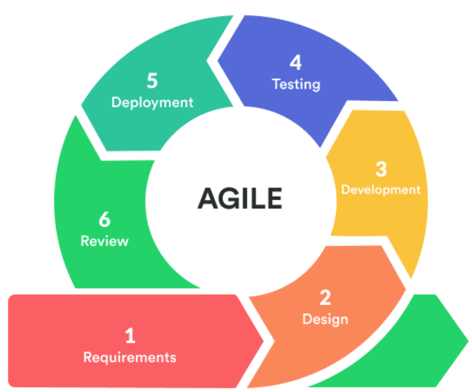

Figure 5: Agile Software Development

The development of the Smart Campus Complaint and Case Management System will follow the Agile Software Development Methodology, specifically utilizing the Scrum framework. Agile was selected for its iterative and incremental approach to software development, which allows the research team to continuously refine system features based on ongoing feedback from stakeholders, including students, faculty, and university administrators. This flexibility is particularly well-suited to the dynamic and evolving requirements of a campus-based information system where user needs and institutional priorities may shift throughout the development process.

The Agile-Scrum approach organizes development into fixed-length iterations called sprints, typically lasting two to four weeks. Each sprint encompasses a defined set of tasks, including planning, design, coding, testing, and review, ensuring that a functional and testable increment of the system is produced at the end of every cycle. This structured yet adaptive process enables the research team to detect and address issues early, minimize development risks, and maintain consistent alignment with the study's objectives.

The development process will be carried out across the following phases:

Phase 1: Requirements Gathering and Planning: In this initial phase, the research team will conduct interviews, surveys, and consultations with key stakeholders at Mapúa University to identify system requirements, user expectations, and operational constraints. The gathered data will be used to define the system's functional and non-functional requirements, establish the product backlog, and outline the overall project scope and timeline.

Phase 2: System Design: Based on the requirements gathered, the team will develop the system architecture, database schema, and user interface prototypes. This phase includes the design of core system components such as the complaint submission module, case tracking dashboard, notification system, and administrative reporting tools. Wireframes and mockups will be presented to stakeholders for validation before development begins.

Phase 3: Iterative Development and Testing: The system will be built incrementally through a series of sprints. Each sprint will focus on the development and testing of specific modules, with regular sprint reviews conducted to evaluate progress and incorporate stakeholder feedback. Unit testing, integration testing, and user acceptance testing (UAT) will be performed throughout this phase to ensure that each component functions as intended and meets the defined requirements.

Phase 4: System Integration and Quality Assurance: Once individual modules have been developed and tested, they will be integrated into a unified system. Comprehensive quality assurance procedures will be conducted to verify the system's overall functionality, performance, security, and usability. Any identified defects or inconsistencies will be resolved prior to deployment.

Phase 5: Deployment and Evaluation: The finalized system will be deployed within the Mapúa University environment for pilot testing. Selected end-users, including students, faculty, and administrators, will interact with the system under real or simulated conditions. Feedback gathered during this phase will be used to evaluate the system's effectiveness in addressing the identified problems and to make final refinements as necessary.

The adoption of the Agile-Scrum methodology ensures that the development of the Smart Campus Complaint and Case Management System remains responsive to stakeholder needs, grounded in iterative improvement, and aligned with the academic and operational goals of Mapúa University.

3.2 System Development Phases

This Smart Campus Complaint & Case Management System for Mapúa University has been divided into six sequential yet connected phases that tackle different parts of the system's lifecycle from inception through continuous management. The combination of these phases provides a structured approach to the development of the system, thorough testing, and deployment that meets the needs of the varied stakeholder community within the university.

A. Planning

Project scope, resource allocation, timelines and requirements are determined for the Smart Campus Complaint & Case Management System. Discussions will be held with Mapúa University stakeholders such as administration, student affairs, and IT support staff to find out existing manual processes and the policy compliance needs. This is when risk assessments are carried out to define a project roadmap and to confirm that the system meets the university's operational guidelines.

B. System Design

The architecture determines the schema, infrastructure, and application interfaces that are implemented on the server side and client side. Wireframes for User Interface (UI) and User Experience (UX) designs are created to facilitate easy navigation for students, faculty, and administration. Component level design is the design of the data flow from the user submitted complaint forms to the backend management system, defines access control for the data contained in the system and identifies the integration points to other campus systems.

C. Development

Frontend Development involves creating user-friendly web interfaces that are responsive for user complaints and engagement with the administrative dashboard. Backend development creates the logic for the server, application programming interfaces (APIs), and safe integrations with the database to maintain user data, log tracking histories, and upload and store multimedia evidence like photos or documents.

D. Predictive Analytics and AI Implementation

Rule-based classification models and predictive algorithms are built and trained to analyze and interpret the description of incoming complaints. The tickets are automatically sorted into different domains of the university (Facilities Management, Academic Affairs, ILMO/IT Support, Student Services, etc.), and a priority severity score is given based on the priority level of the text. This is when administrators review historical data on campus complaints and create predictive campus dashboards to show trends that create delays or campus problems that recur seasonally.

E. Testing

Each unit is tested to validate individual units, like the status update workflow or filing form. Integration testing ensures the predictive classification engine's ability to communicate with the core database seamlessly. System and User Acceptance Testing (UAT) is performed with a small group of Mapúa students, faculty, staff, and officials to check for usability and data confidentiality and to address performance issues prior to official launch.

F. Deployment and Maintenance

The finalized system is deployed to production servers or university cloud infrastructure, to be used actively on the campus. After deployment, the maintenance phase starts, during which the system is monitored continuously, server health checks are conducted, and data backup procedures are implemented. This involves periodic software patches to appease customer input, fix probable bugs, streamline database queries with time as data accumulates, and refine predictive classification models to keep them accurate over time.

3.3 Research Methodology

A. Research Design

The study employs a mixed-methods research design to support the development and evaluation of the proposed web-based complaint and case management system for Mapúa University. This approach integrates both qualitative and quantitative methods, enabling the researchers to gather comprehensive insights that neither method could fully provide on its own. A mixed-methods design is particularly suited to studies that aim to both understand a problem in depth and measure outcomes systematically, as it combines the explanatory strength of qualitative data with the generalizability of quantitative data (Creswell & Creswell, 2018).

The qualitative component focuses on understanding users' experiences, challenges, and suggestions related to the existing complaint-handling and administrative processes within Mapúa University. This component draws primarily from interviews with key informants, such as the Student Success Coordinator, whose firsthand accounts help establish how concerns are currently received, routed, and resolved. These insights inform the design of the proposed system and the formulation of its features.

The quantitative component measures user perceptions through structured Likert-scale questions, particularly in terms of the system's usability, functionality, and efficiency. The use of Likert-scale instruments allows respondents' attitudes and perceptions to be quantified and analyzed using descriptive statistics, providing a measurable basis for evaluation (Joshi et al., 2015).

By combining these two components, the research design allows the researchers to evaluate both the functional aspects of the proposed system and the perceptions of its users in a structured and measurable manner. This integration ensures that the development of the system is guided not only by numerical evaluation but also by the real experiences and needs of its intended users, resulting in a more holistic and user-centered assessment.

B. Research Locale

The study is conducted within Mapúa University, focusing on its academic and administrative environment. Data gathering takes place among students and university administrators who are directly involved in or affected by university-related processes such as academic records management, administrative transactions, and institutional services.

The selected locale is appropriate for the study as it provides direct access to the target users of the proposed system, allowing the researchers to gather relevant and practical feedback regarding system requirements and usability within a real academic setting

C. Research Respondents

The study targets a total of 30 respondents, consisting of individuals directly involved with Mapúa University. This includes students and university administrators who are engaged in various academic and administrative processes within the institution.

Among these respondents are two key informants who were interviewed regarding the existing complaint-handling process: Ms. Gemini Magana, the Student Success Coordinator of the Institute for Digital Learning, and a Life Coach who assists the university's online students. As personnel who regularly receive, review, and route student concerns, they provide firsthand insight into how complaints are currently submitted, prioritized, tracked, and resolved under the present manual setup, which directly informs the design and evaluation of the proposed system.

The respondents are selected to represent both end-users and system-related stakeholders to ensure that different perspectives are considered in evaluating the proposed web-based management system. Students provide insights based on their experience with academic requirements, enrollment processes, and other student services, while administrators and student support personnel, such as coordinators and life coaches, contribute perspectives related to complaint handling, documentation, and institutional workflow management.

The selection of these respondents ensures that the study gathers balanced and relevant feedback from users who are directly affected by and involved in the university's operational processes, allowing for a more comprehensive evaluation of the proposed system.

D. Research Instruments

The study utilizes structured questionnaires as the primary research instrument for data collection, administered through Google Forms. The instrument is designed to collect both qualitative and quantitative data aligned with the objectives of the study.

The questionnaire consists of the following components:

- Open-ended questions, which allow respondents to express their experiences, challenges, and suggestions regarding existing academic and administrative processes within Mapúa University.

- Likert scale questions, which measure respondents' level of agreement on statements related to system usability, efficiency, and overall satisfaction with the proposed web-based system.

These instruments are designed to ensure that both descriptive insights and measurable data are collected, providing a comprehensive basis for evaluating the proposed system's potential impact on university operations.

E. Data Gathering Procedure

The data-gathering process for this study will be carried out in three phases. In the first phase, the researchers will seek formal approval from the university administration and obtain informed consent from all target respondents prior to survey distribution. The structured questionnaire will then be distributed digitally through Google Forms to the identified respondents, consisting of students and university administrators of Mapúa University. Respondents will be given a designated period to complete the survey at their convenience to minimize disruption to academic and administrative activities.

In the second phase, the researchers will conduct a pilot testing session in which selected end-users interact with a working prototype or deployed version of the Smart Campus Complaint and Case Management System. This session will provide respondents with direct experience of the system's core features, including complaint submission, real-time status tracking, in-system notifications, and the administrative dashboard. Following the session, respondents will be asked to complete the structured questionnaire to capture their immediate evaluations of system usability, functionality, and efficiency.

In the third phase, all completed survey responses will be collected and compiled for analysis. The researchers will review the gathered data for completeness and consistency before proceeding to the data analysis stage. Any incomplete or invalid responses will be excluded to maintain the integrity of the dataset.

F. Data Analysis Procedure

The data collected through the structured questionnaire will be analyzed using both qualitative and quantitative methods, consistent with the mixed-methods research design of the study. Quantitative data derived from the Likert scale items will be tabulated and subjected to descriptive statistical analysis, including the computation of frequency counts, percentages, weighted means, and overall mean scores for each evaluation criterion. These results will be used to measure respondents' perceptions of the system in terms of usability, functionality, and efficiency across its core features.

Qualitative data gathered from the open-ended questions will be analyzed through thematic analysis. Responses will be reviewed, coded, and grouped into recurring themes that reflect the experiences, challenges, and recommendations shared by respondents regarding both the current manual complaint process and the proposed system. These themes will be used to complement and contextualize the quantitative findings, providing a richer understanding of user needs and system impact.

The integrated analysis of quantitative and qualitative results will serve as the basis for drawing conclusions about the system's overall effectiveness and for identifying areas for improvement. All findings will be presented in narrative and tabular forms within the results and discussion chapter of the study.

G. Ethical Considerations

The conduct of this study is guided by ethical principles to ensure the protection, dignity, and rights of all research participants. Prior to data collection, informed consent will be obtained from all respondents. Each participant will be clearly informed of the purpose of the study, the nature of their involvement, and their right to withdraw at any time without consequence. Participation in the study is entirely voluntary, and no respondent will be subjected to any form of coercion or undue influence to participate.

Confidentiality and anonymity of respondents will be strictly maintained throughout the study. All data collected will be used solely for academic research purposes and will not be shared with third parties or used beyond the scope of this study. Respondents' personal information will not be disclosed in any publication or presentation arising from this research. Survey responses will be stored securely and accessible only to the members of the research team.

The researchers also commit to the honest and accurate reporting of all findings. Data will not be manipulated, selectively omitted, or misrepresented to favor a particular outcome. Any limitations encountered during the study will be transparently acknowledged. The research team will likewise ensure that the system itself is designed with user privacy in mind, with complaint data accessible only to authorized personnel and protected through appropriate security measures in accordance with the university's data governance policies.

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Appendices:

A. **Gemini Magana**

1. May I know your name and position at Mapúa University? Single-line text.

“Gemini Magana—Student Success Coordinator of the Institute for Digital Learning."

Section A — Intake: how a complaint enters the system

2. When a student has a complaint or concern, where do they go first? Is there one office, or does it depend on the type of issue?

"I think it depends on the type of issue the student is experiencing. For example, when it comes to course-related concerns, students often reach out to me when they have not received a response from their facilitator or when they need additional clarification about course requirements, deadlines, or learning activities.”

3. What are the actual ways a student can submit a complaint right now (in person, email, a paper form, a phone call, a Facebook page, a suggestion box, or a specific office)?

"If students need to contact me directly, I prefer that they message me through Microsoft Teams, as it allows me to respond more efficiently and keep track of our conversations."

4. Is there an official form a student fills out, or is it informal? If there's a form, is it paper or digital?

"There is no specific form that students need to complete to contact me. They can simply reach out to me directly through Microsoft Teams, which is my preferred method of communication because it allows for quicker responses and easier follow-up. Students may also contact me via Outlook email if that is more convenient for them.”

5. Does the student need to identify themselves, or can complaints be submitted anonymously?

“To help me address their concerns thoroughly and effectively, students should identify themselves and provide their full name, course, and section when reaching out. They should also clearly explain the concern or issue they are experiencing, including any relevant details that may help me better understand the situation."

6. Roughly how many complaints does your office receive in a typical week or month?

"On average, I would estimate that I handle at least 30 complaints or concerns from students. These concerns vary in nature and may include course-related issues, communication concerns, technical difficulties, or requests for academic support."

Section B — Routing: how it reaches the right people

7. Once a complaint comes in, who first reads or receives it?

"As the Student Success Coordinator, I am primarily responsible for receiving, reviewing, and addressing student complaints and concerns. It is my role to listen to their concerns, assess the situation, provide guidance when appropriate, and coordinate with the relevant faculty, facilitators, or departments to help resolve issues effectively."

8. How do you decide which office or person should handle a particular complaint? Is there a written rule, or is it based on experience?

"The office or person responsible for handling a particular complaint is determined based on the nature and subject of the concern. While there may be established institutional guidelines regarding the roles and responsibilities of different offices, my decision is often guided by both these guidelines and my experience in handling student concerns."

9. What are the main categories of complaints you receive (facilities, academic, IT, enrollment, conduct, etc.)?

" Students typically reach out to me regarding a variety of issues, including course-related concerns, communication challenges, academic support needs, and other matters that may affect their success in the program."

10. What happens when a complaint involves more than one office or when it's unclear who's responsible?

"When a complaint involves more than one office or when it is not immediately clear who is responsible, I begin by reviewing the details of the concern and identifying all the areas involved. I then coordinate with the relevant offices or personnel to ensure that the complaint is directed to the appropriate individuals for review and resolution."

11. How is the complaint physically or digitally passed along—email, forwarded paper, a logbook, or a shared spreadsheet?

"The method of forwarding a complaint depends on the nature of the concern. For course-related issues, complaints are typically handled and referred digitally through email and/or Microsoft Teams. However, when the concern involves offices such as the Treasury or the Registrar, I usually address the matter personally by visiting the respective office. This approach allows me to discuss the concern directly with the responsible personnel, clarify any questions immediately, and facilitate a faster resolution when necessary."

Section C — Processing and approval

12. After a complaint is assigned, what are the typical steps taken to resolve it?

"After a complaint is assigned to the appropriate office or person, the first step is usually to review the details and gather any necessary information related to the concern. The responsible office or personnel will assess the issue, verify the facts, and determine the appropriate course of action."

13. Do certain complaints need approval from a higher authority (a dean, director, or department head) before action is taken? Which kinds?

"Yes. In most cases, complaints are initially handled by the appropriate staff member or office responsible for the concern. However, if the issue cannot be resolved at that level, it may be escalated to a higher authority, such as a department head, director, dean, or other designated administrator for further review and action."

14. Who is responsible for actually following up and making sure something gets done?

"As the student success coordinator, I am responsible for following up on student complaints and ensuring that appropriate action is taken. Once a concern has been referred to the relevant office or personnel, I monitor its progress, communicate with the individuals involved, and follow up as needed to ensure that the issue is being addressed in a timely manner."

15. Are there complaints that tend to get stuck or delayed? What usually causes the delay?

"Yes, there are some complaints that may experience delays in the resolution process. In many cases, the delay occurs because the concern was overlooked, missed during communication, or requires follow-up from multiple individuals or offices. Delays can also happen when the responsible person is waiting for additional information, supporting documents, or a response from another department before taking action."

Section D — Tracking and records

16. How do you keep track of a complaint while it's being worked on? Is there a logbook, spreadsheet, email thread, or filing system?

"I keep track of student complaints using both email threads. Email threads allow me to maintain a complete record of communications, including the concern raised, responses from the involved offices or personnel, and any updates regarding the case."

17. Can you tell at any moment what the current status of a complaint is (e.g., new, in progress, resolved)? How?

"No."

18. After a complaint is resolved, where is the record kept, and for how long?.

"Once a complaint has been resolved, there is no formal long-term record retention process in place. The communication related to the concern may remain within the existing email thread or tracking spreadsheet while the issue is being addressed, but resolved complaints are not typically archived or maintained as permanent records."

19. Are past complaints reviewed to spot recurring problems, or is each one handled on its own?

“Yes, past complaints are reviewed to identify recurring issues or patterns. While each complaint is handled individually to ensure that the student's specific concern is properly addressed, reviewing past complaints helps determine whether similar concerns are being raised repeatedly.”

Section E — Complaint Prioritization

20. When several complaints come in at the same time, how do you decide which one to handle first? Is there a system for ranking urgency, or is it based on judgment?

“When multiple complaints are received at the same time, I generally handle them on a first-come, first-served basis. The order of priority is typically determined by who contacted me first, whether through Microsoft Teams or Outlook email. However, I still review each concern upon receipt to understand its nature and ensure it is directed to the appropriate office or personnel. By following the order in which complaints are received, I can manage concerns fairly and systematically while ensuring that all students receive timely attention and support.”

21. Are some categories of complaints automatically treated as more urgent than others — for example, is a broken air-conditioning unit handled faster than a grade dispute, or a safety issue faster than a facilities request? What makes a complaint "high priority" in your office — is it the type of issue, how many students it affects, a deadline involved, who submitted it, or something else?

"The order of priority is typically determined by who contacted me first, whether through Microsoft Teams or Outlook email. By following the order in which complaints are received, I can manage concerns fairly and systematically while ensuring that all students receive timely attention and support.”

22. Is there any official policy or guideline that tells staff how to rank complaints, or does each person decide on their own?

“None.”

23. How do you handle a complaint that seems urgent to the student but is actually routine from your side (or the reverse)?

“When handling complaints, I understand that the level of urgency perceived by the student may sometimes differ from how the concern is classified from an operational perspective. If a student considers a complaint urgent, I make it a priority to listen carefully, acknowledge their concern, and provide reassurance that the matter will be reviewed and addressed appropriately.”

Section F — Communication with the student

24. How does the student find out their complaint was received?

“Students are informed that their complaint has been received through direct communication from them or from the concerned department, either via Microsoft Teams or Outlook email.”

25. Are students updated on the progress of their complaint? If so, how and how often?

“Students do not typically receive regular updates from the office handling their complaint. As the Student Success Coordinator, I take the initiative to follow up on pending concerns and remind the responsible office or personnel about the complaint when necessary.”

26. How does the student learn that their complaint has been resolved?

“Students are informed that their complaint has been resolved through direct communication from me or from the concerned department, either via Microsoft Teams or Outlook email.”

27. If you need more information from the student, how do you reach them?

“If I need additional information or clarification regarding a complaint, I usually contact the student directly through Microsoft Teams. This is my primary method of communication because it allows for quick and efficient exchanges and makes it easier to follow up on outstanding concerns.”

Section G — Pain points

28. What are the biggest difficulties your office faces with the current way of handling complaints?

“One of the biggest difficulties in our current complaint-handling process is delays in responses and resolution from other offices involved in addressing the concern. While complaints can be received, reviewed, and forwarded promptly, the resolution process may take longer when action or feedback is needed from another department.”

29. Have complaints ever been lost, forgotten, or duplicated? What happened?

"No."

30. From your experience, what do students complain about most regarding the complaint process itself?

“Delay”

31. If you could change one thing about how complaints are handled today, what would it be?

"Based on my experience, the most common complaint students have about the complaint-handling process is the delay in receiving responses and resolutions. Students often expect their concerns to be addressed quickly, especially when the issue affects their coursework, enrollment, grades, or access to services.”

32. What features would you most want in a digital complaint system, if one existed?

“If a digital complaint system were available, I would want features that improve tracking, communication, and accountability throughout the complaint resolution process. One of the most important features would be the ability to track the status of complaints in real time, allowing both students and staff to see whether a complaint has been received, assigned, being reviewed, or resolved.”

B. Leanna Mae A. Cruz

1. May I know your name and position at Mapúa University?

“Life coach”

Section A — Intake: how a complaint enters the system

2. When a student has a complaint or concern, where do they go first? Is there one office, or does it depend on the type of issue?

“If a student’s concern exceeds our knowledge, we recommend they message a specific person. For instance, if they have a concern with a professor, we advise them to message the professor and include us in the email.”

3. What are the actual ways a student can submit a complaint right now (in person, email, a paper form, a phone call, a Facebook page, a suggestion box, a specific office)?

“Since we work with online students, we ask them to email their concerns so we can follow up.”

4. Is there an official form a student fills out, or is it informal? If there's a form, is it paper or digital?

“It's just an informal message.”

5. Does the student need to identify themselves, or can complaints be submitted anonymously?

“They identify themselves since it was an email."

6. Roughly how many complaints does your office receive in a typical week or month?

“More than 20 complaints per month”

Section B — Routing: how it reaches the right people

7. Once a complaint comes in, who first reads or receives it?

“This is a personal message. I am the only one who can received the message."

8. How do you decide which office or person should handle a particular complaint? Is there a written rule, or is it based on experience?.

“It’s based on personal experience. As a life coach, I must understand both perspectives. I must remain objective when providing advice to avoid any bias.”

9. What are the main categories of complaints you receive (facilities, academic, IT, enrollment, conduct, etc.)?

“More likely, enrollments, academic concerns, IT concerns and registrar concerns."

10. What happens when a complaint involves more than one office or when it's unclear who's responsible?

“Since there’s an assigned department for UOX, we refer them to IDL to resolve any difficulty problems they may encounter.”

11. How is the complaint physically or digitally passed along—email, forwarded paper, a logbook, a shared spreadsheet?

“We passed it via email and MS Teams.”

Section C — Processing and approval

12. After a complaint is assigned, what are the typical steps taken to resolve it?

“Sometimes, when students receive the answers they needed, it doesn’t necessarily mean that follow-ups were requested. In such cases, it’s considered a resolved concern.”

13. Do certain complaints need approval from a higher authority (a dean, director, or department head) before action is taken? Which kinds?

“No need for approval, but sometimes there are cases where the concerns are still pending and not answered by a person needed. We tell the students to include the heads.”

14. Who is responsible for actually following up and making sure something gets done?

“We are the one who is responsible for follow-ups and updates."

15. Are there complaints that tend to get stuck or delayed? What usually causes the delay?

"Yes, those are faculty concerns, such as student activity or those who have delayed their tasks."

Section D — Tracking and records

16. How do you keep track of a complaint while it's being worked on? Is there a logbook, spreadsheet, email thread, or filing system?

"We keep track via email thread."

17. Can you tell at any moment what the current status of a complaint is (e.g., new, in progress, resolved)? How?

"We mark the concerns as resolved, and some are still pending, but we make sure all the concerns must be resolved."

18. After a complaint is resolved, where is the record kept, and for how long?

"Yes, we cannot disclose this one."

19. Are past complaints reviewed to spot recurring problems, or is each one handled on its own?

"Most of the problems are recurring, and still we find ways to solve them."

Section E — Complaint Prioritization

20. When several complaints come in at the same time, how do you decide which one to handle first? Is there a system for ranking urgency, or is it based on judgment?

"I prioritize complaints based on urgency and impact. Critical issues are handled first, while less urgent concerns are addressed in order. I also use good judgment to ensure every customer receives timely attention and updates."

21. Are some categories of complaints automatically treated as more urgent than others—for example, is a broken air-conditioning unit handled faster than a grade dispute or a safety issue

handled faster than a facilities request? What makes a complaint "high priority" in your office—is it the type of issue, how many students it affects, a deadline involved, who submitted it, or something else?

"Yes. High-priority complaints are usually those involving safety, urgent facilities issues, or matters that affect many students. We prioritize based on the urgency and impact of the issue, not on who submitted the complaint."

22. Is there any official policy or guideline that tells staff how to rank complaints, or does each person decide on their own?

"There are guidelines for handling complaints, but staff also use their judgment to assess the urgency and impact of each case."

23. How do you handle a complaint that seems urgent to the student but is actually routine from your side (or the reverse)?

"I explain the process clearly, assess the complaint based on its urgency and impact, and ensure the student is informed while handling the concern according to the proper priority."

Section F — Communication with the student

24. How does the student find out their complaint was received?

"The student is usually informed through an acknowledgment, such as an email, message, or official receipt, confirming that the complaint has been received and is being processed."

25. Are students updated on the progress of their complaint? If so, how and how often?

"Yes. Students are updated through email, phone, or official messages, especially when there are significant developments or once a resolution has been reached."

26. How does the student learn that their complaint has been resolved?

"The student is informed through an official email, message, phone call, or written notice once the complaint has been resolved."

27. If you need more information from the student, how do you reach them?

“We reached them via email or MS Teams."

Section G — Pain points

28. What are the biggest difficulties your office faces with the current way of handling complaints?

“More likely the faculty."

29. Have complaints ever been lost, forgotten, or duplicated? What happened?

“We follow them up."

30. From your experience, what do students complain about most regarding the complaint process itself?

“About their activities”

31. If you could change one thing about how complaints are handled today, what would it be?

“I would improve the complaint tracking system so students can easily monitor the status of their complaints and receive timely updates.”

32. What features would you most want in a digital complaint system, if one existed?

“I would want features such as online complaint submission, real-time status tracking, automatic updates, and secure communication between students and staff.”